

BFS Traversal

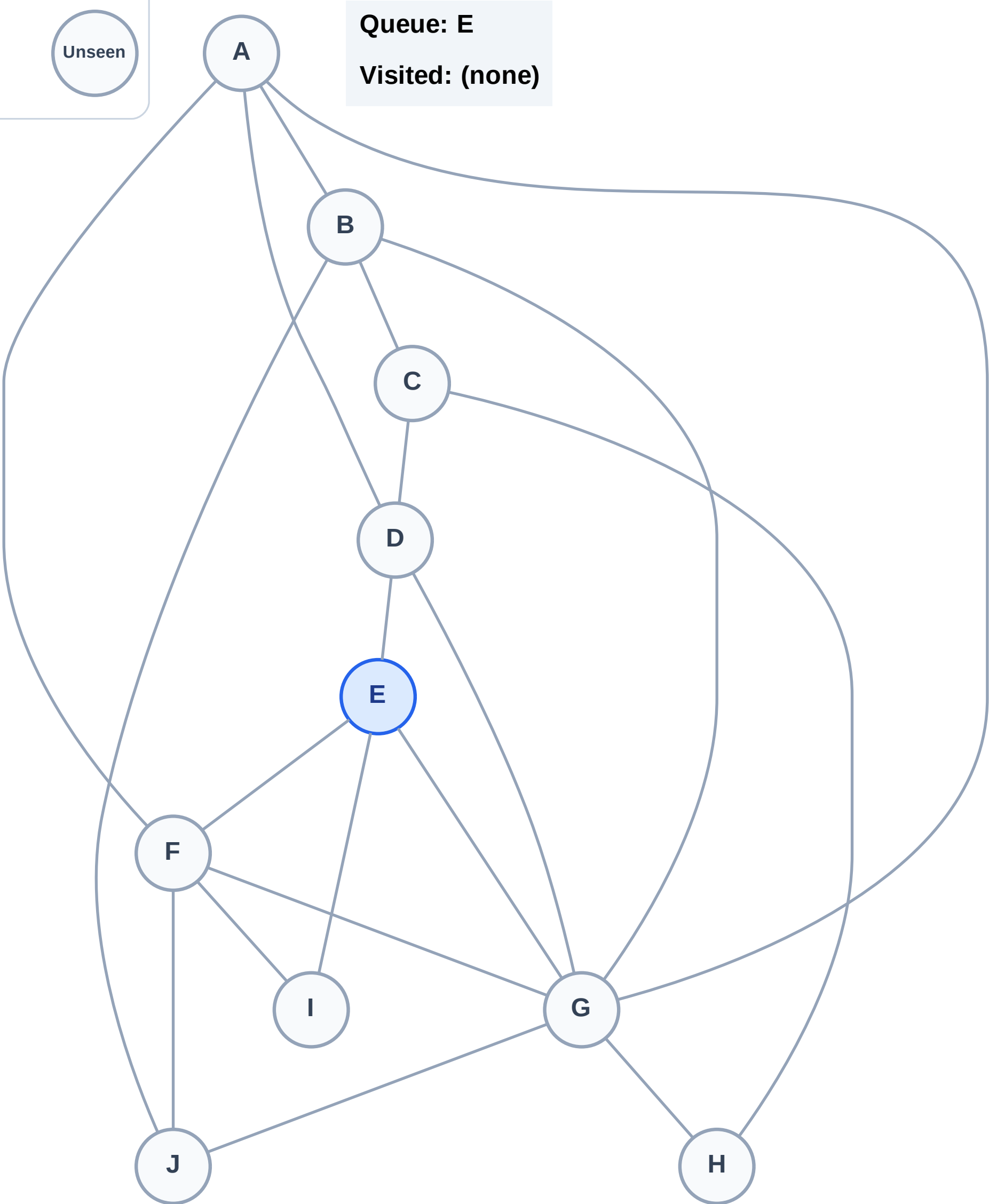
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

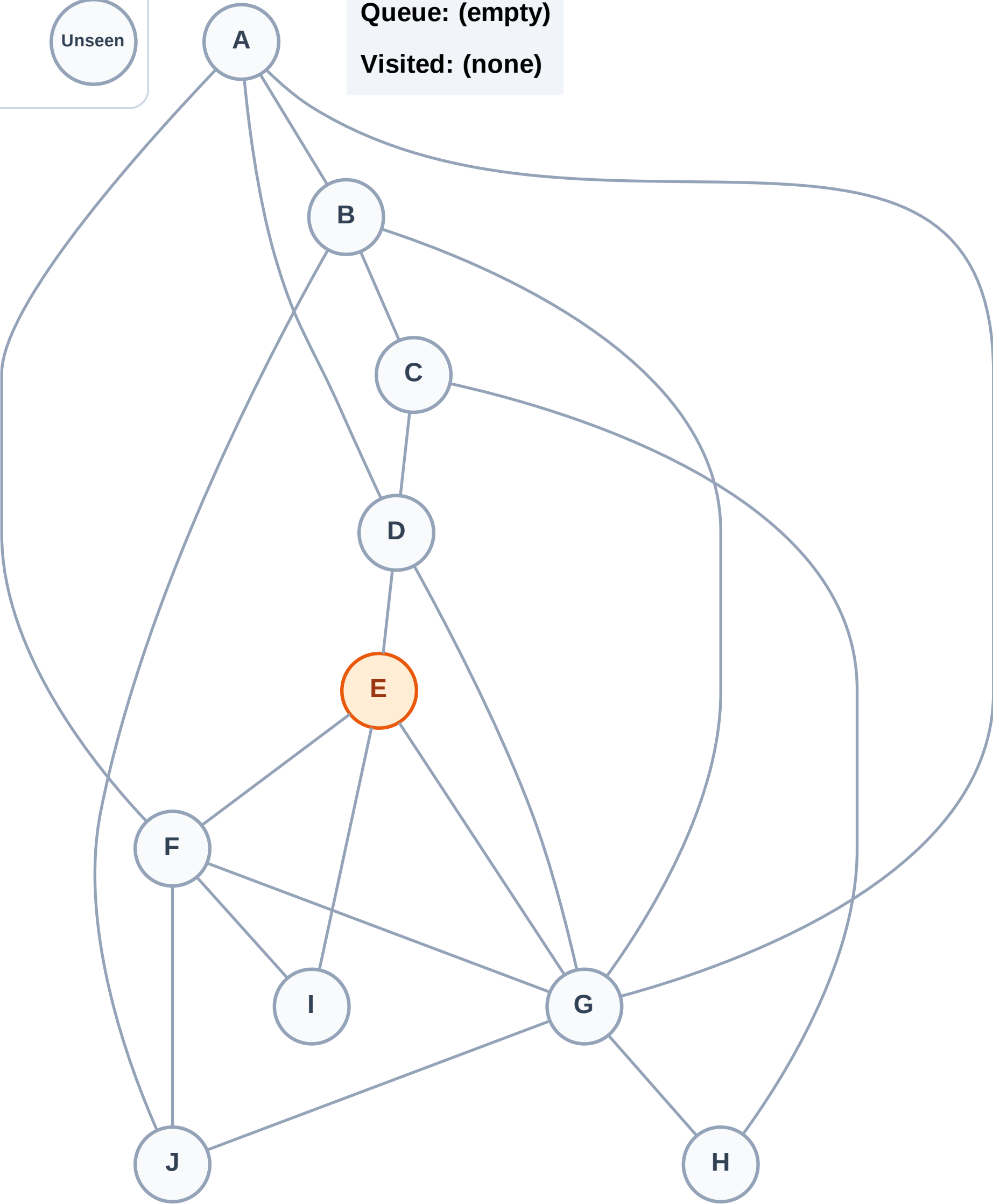
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

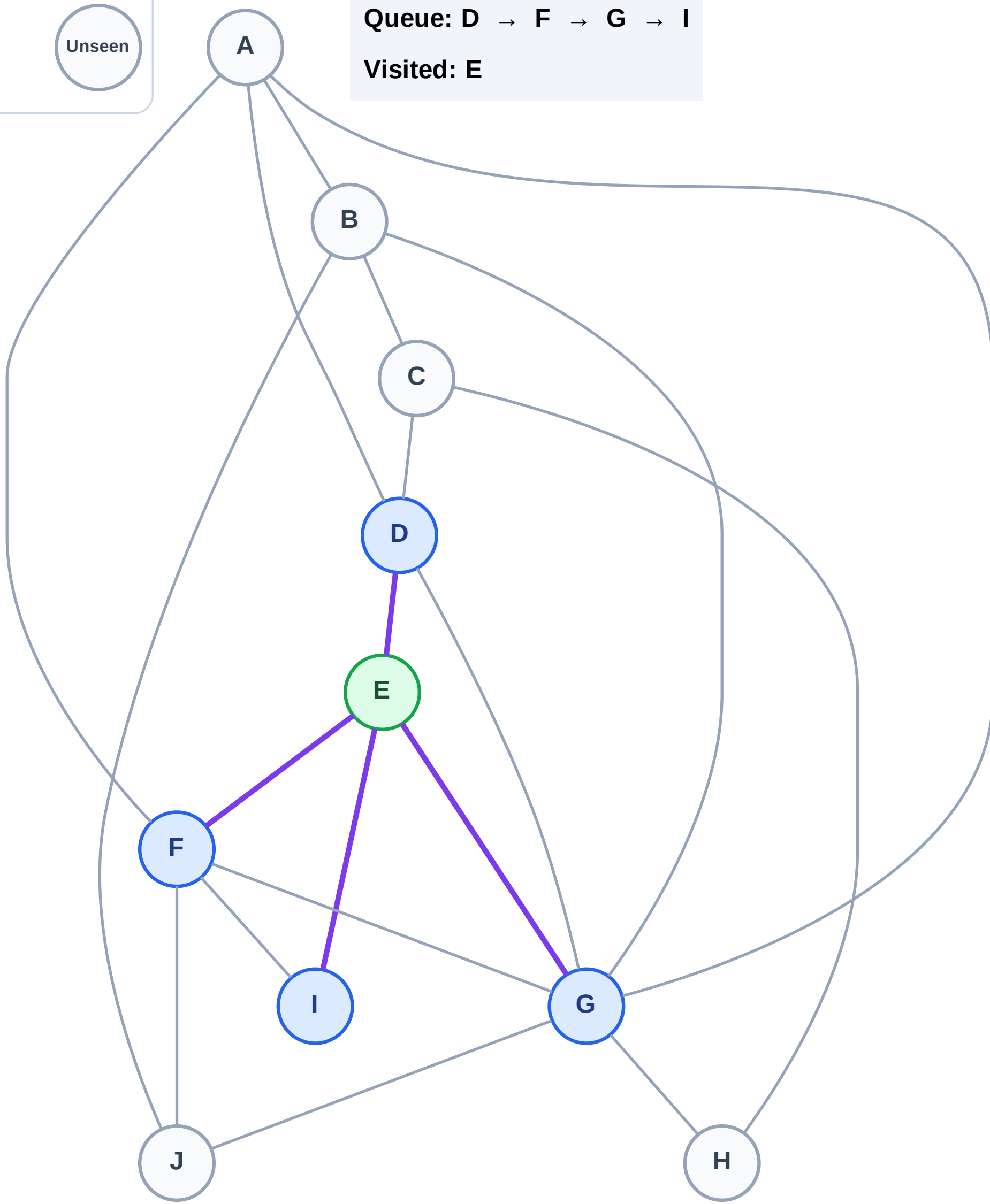
Step 3: Discover D, F, G, I from E

Legend



Queue: D → F → G → I

Visited: E



BFS Traversal

Step 4: Process node D

Legend

Complete

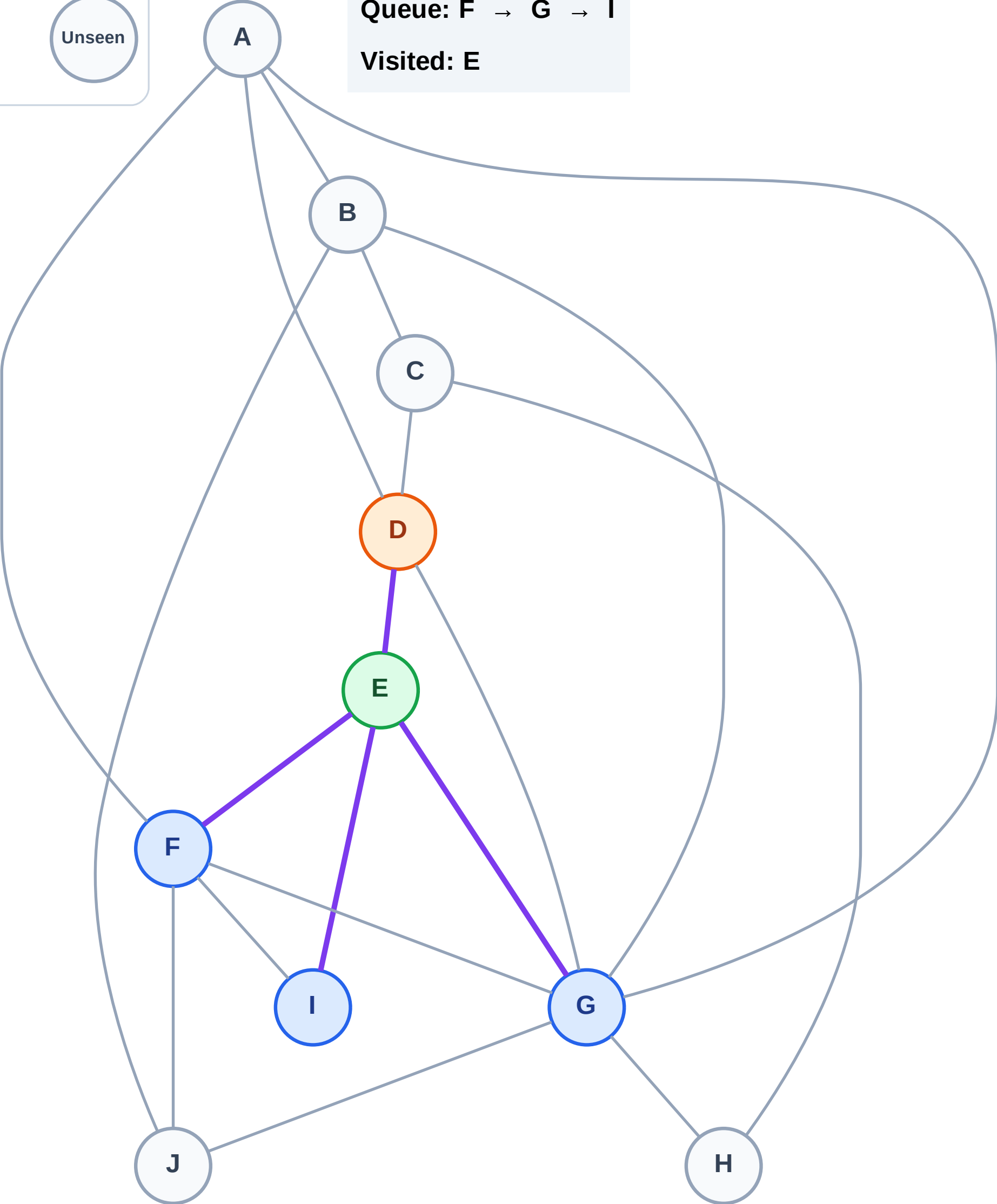
Processing

Discovered

Unseen

Queue: F → G → I

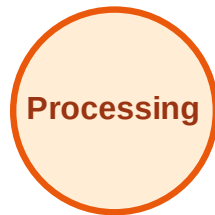
Visited: E



BFS Traversal

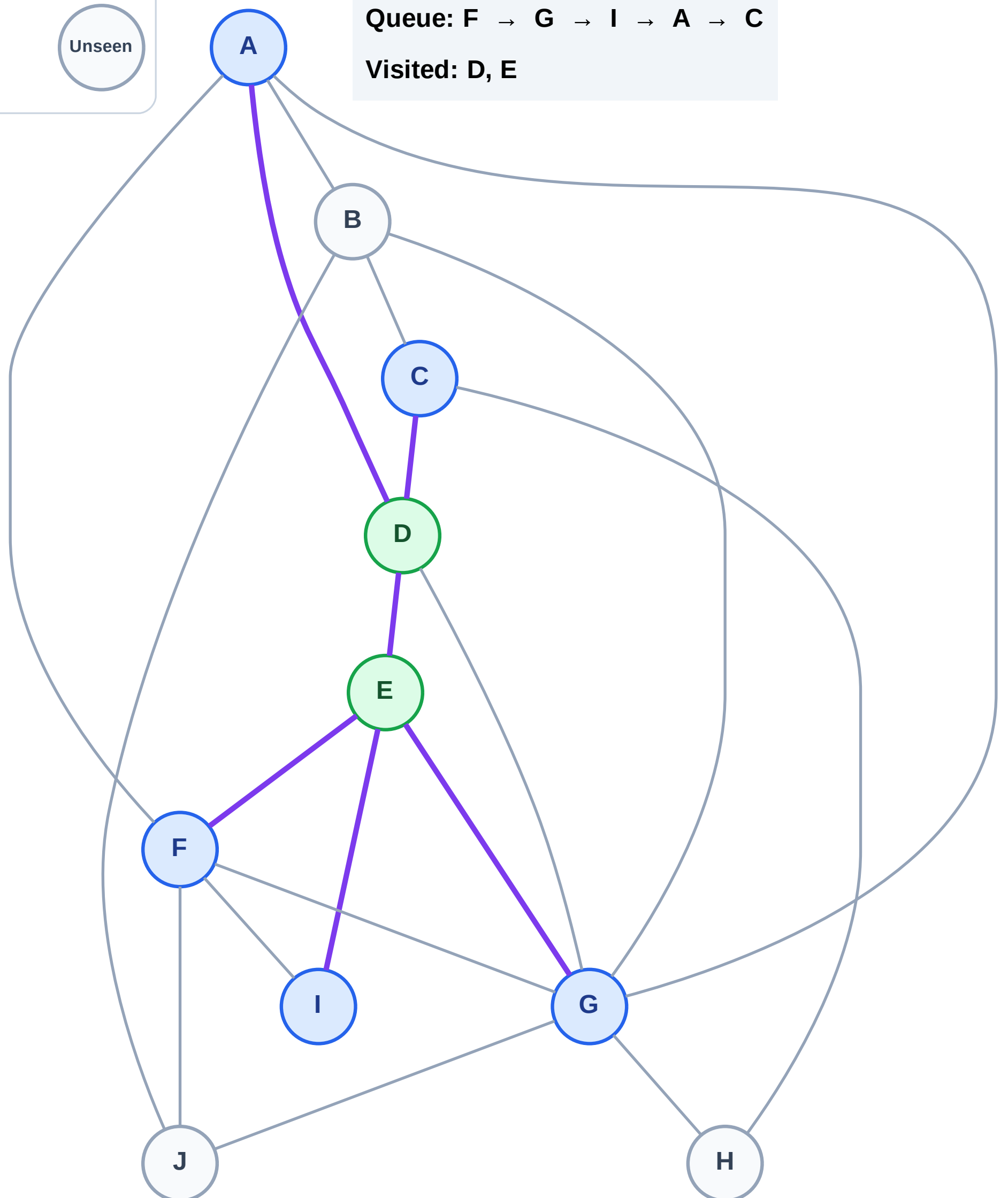
Step 5: Discover A, C from D

Legend



Queue: F → G → I → A → C

Visited: D, E



BFS Traversal

Step 6: Process node F

Legend

Complete

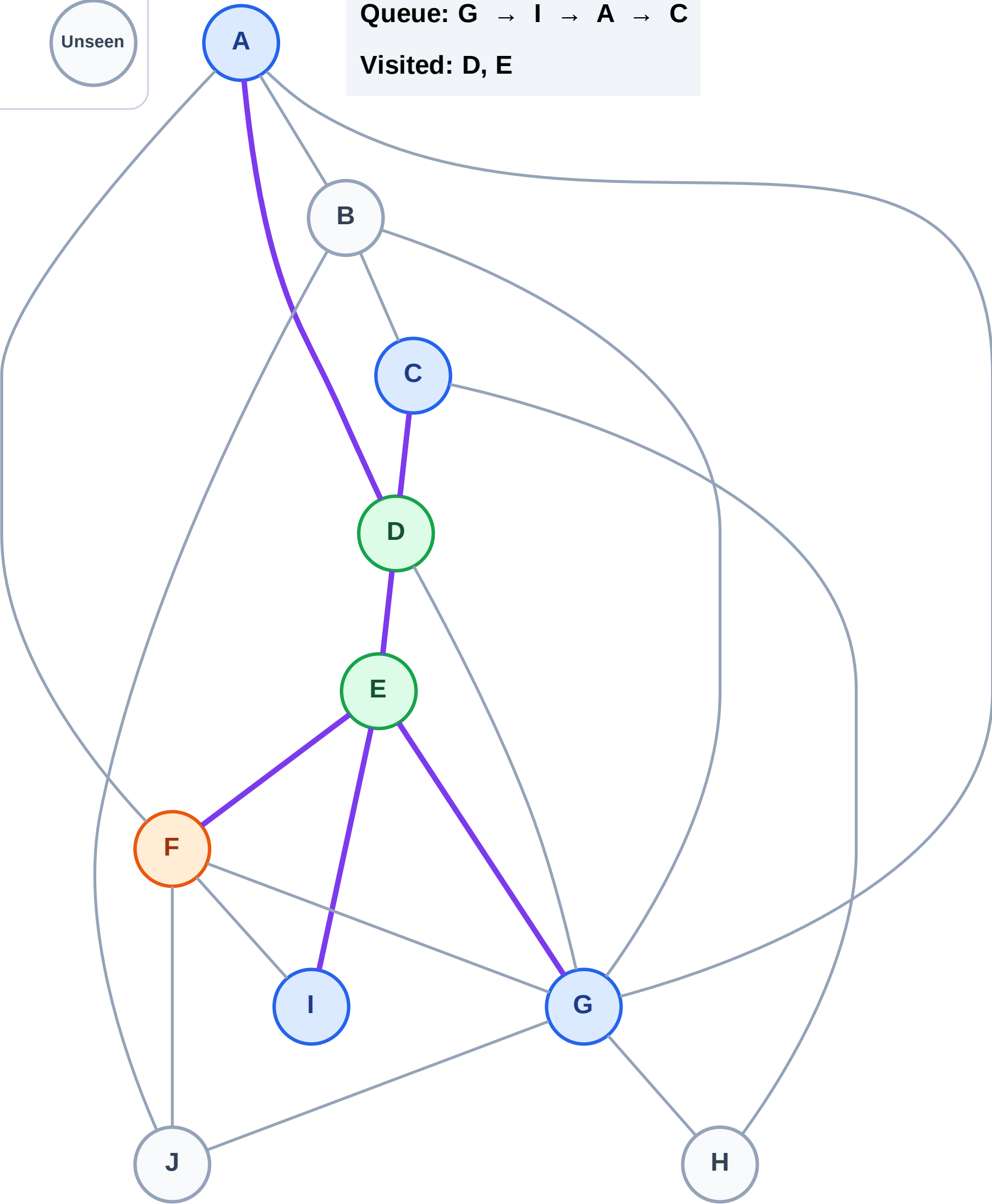
Processing

Discovered

Unseen

Queue: G → I → A → C

Visited: D, E



BFS Traversal

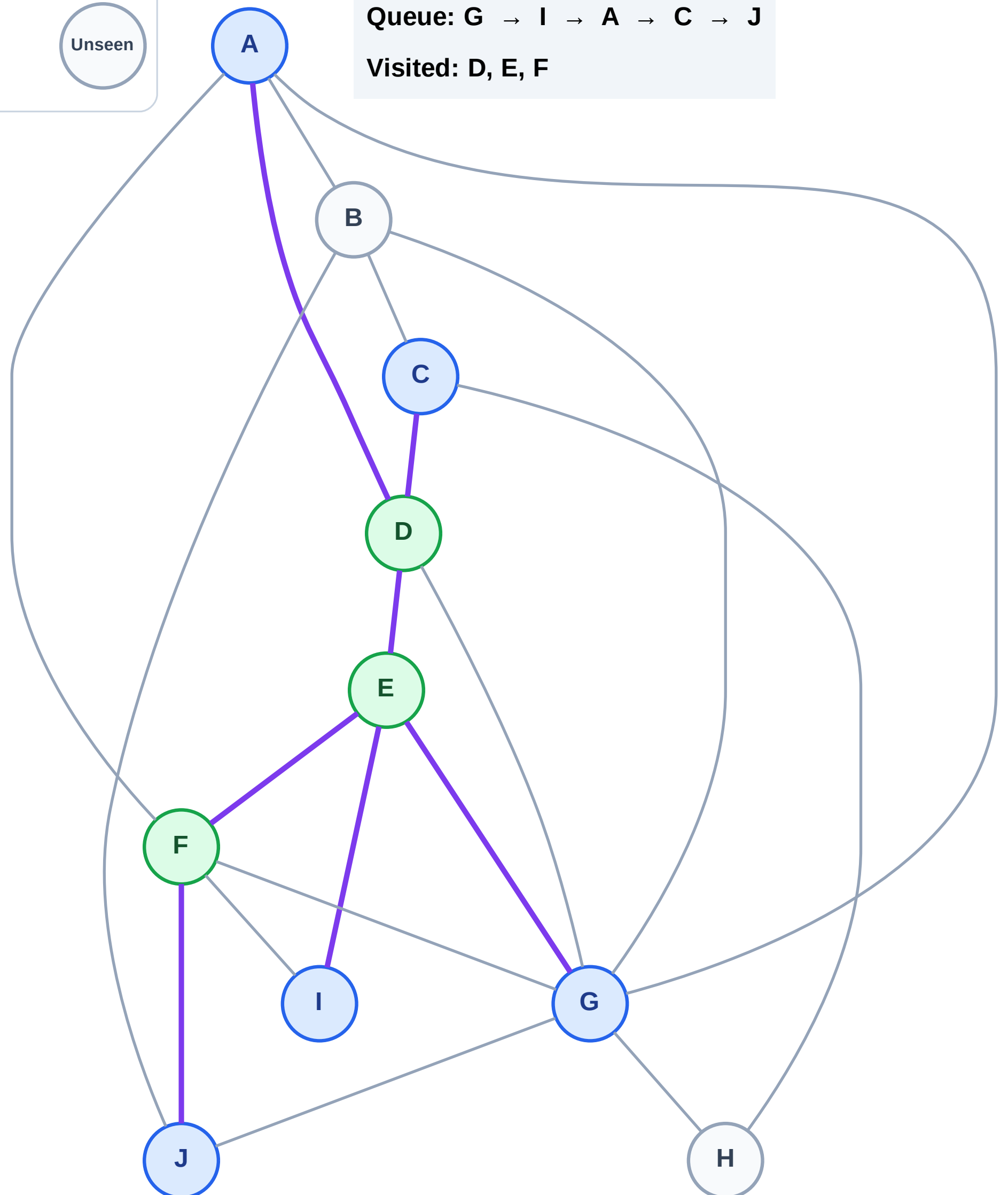
Step 7: Discover J from F

Legend



Queue: G → I → A → C → J

Visited: D, E, F



BFS Traversal

Step 8: Process node G

Legend

Complete

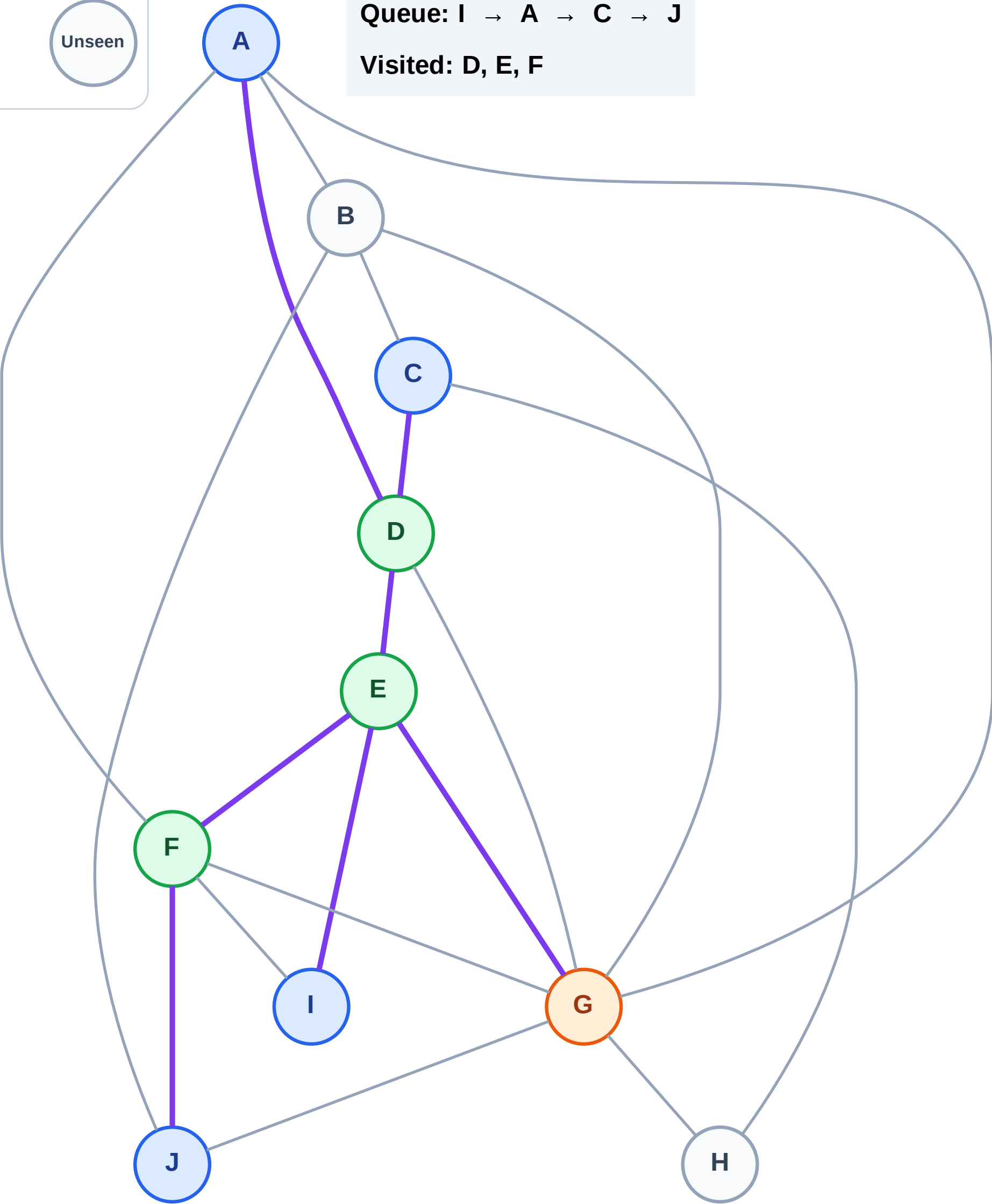
Processing

Discovered

Unseen

Queue: I → A → C → J

Visited: D, E, F



BFS Traversal

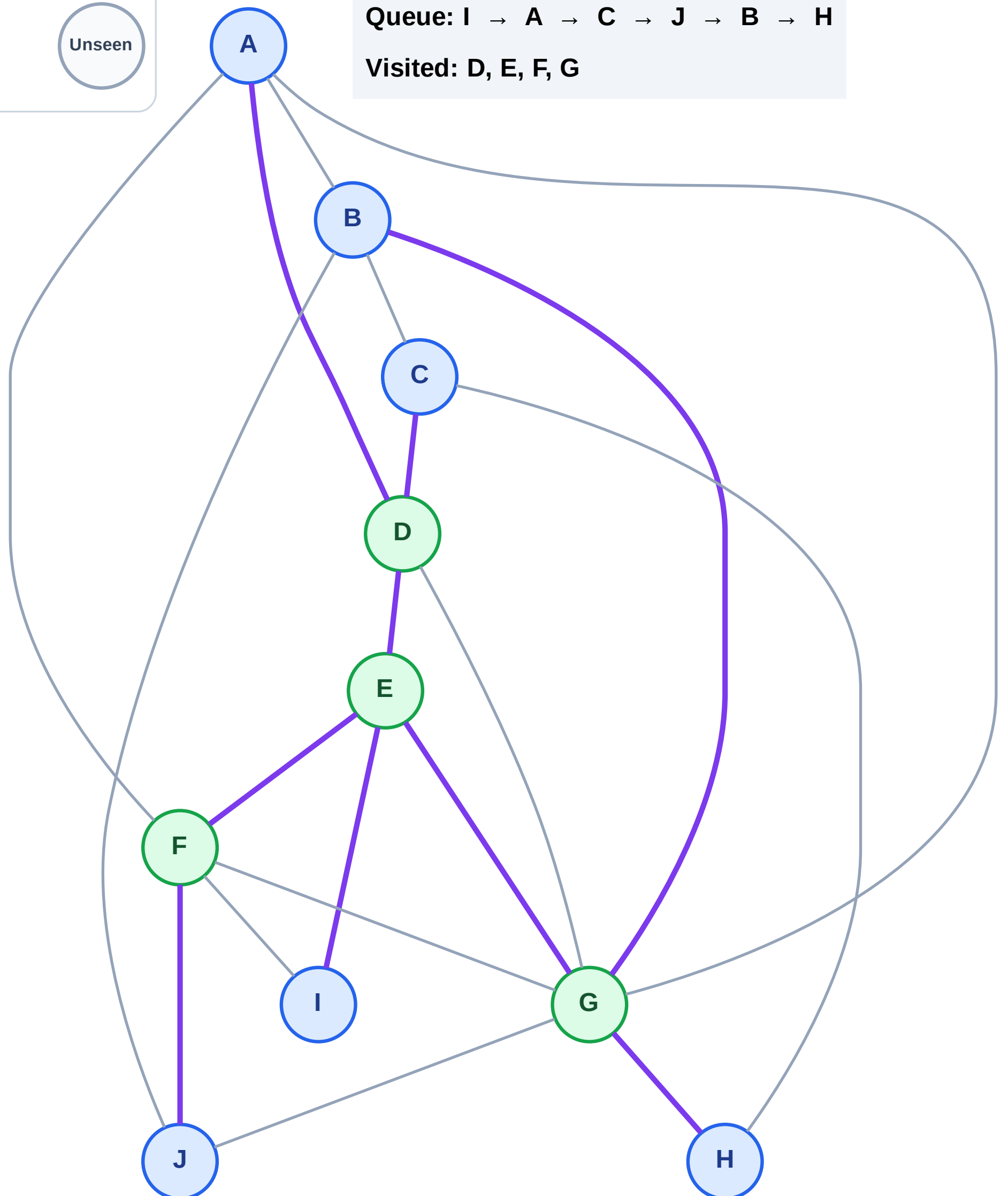
Step 9: Discover B, H from G

Legend



Queue: I → A → C → J → B → H

Visited: D, E, F, G



BFS Traversal

Step 10: Process node I

Legend

Complete

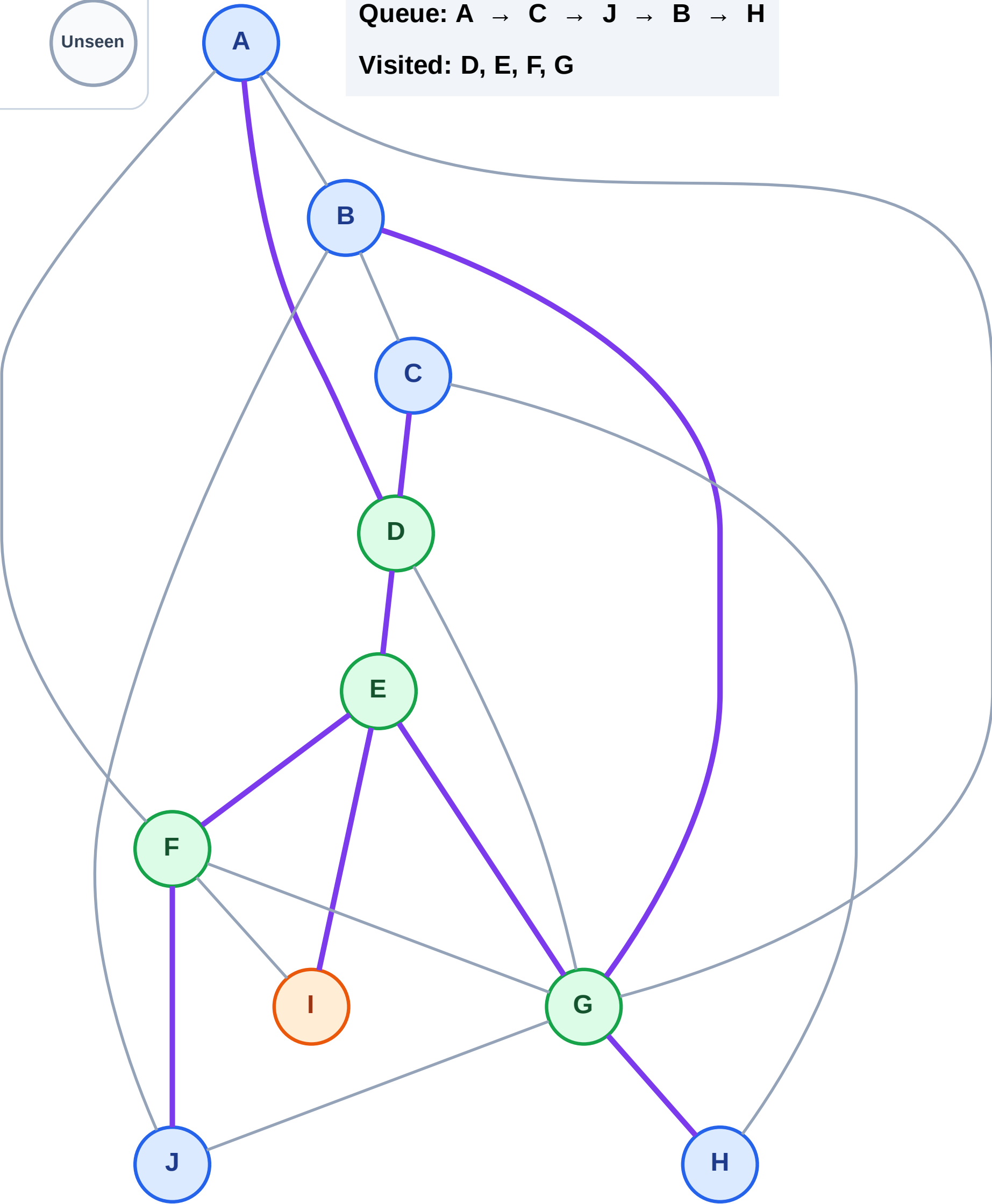
Processing

Discovered

Unseen

Queue: A → C → J → B → H

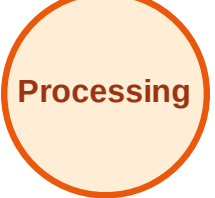
Visited: D, E, F, G



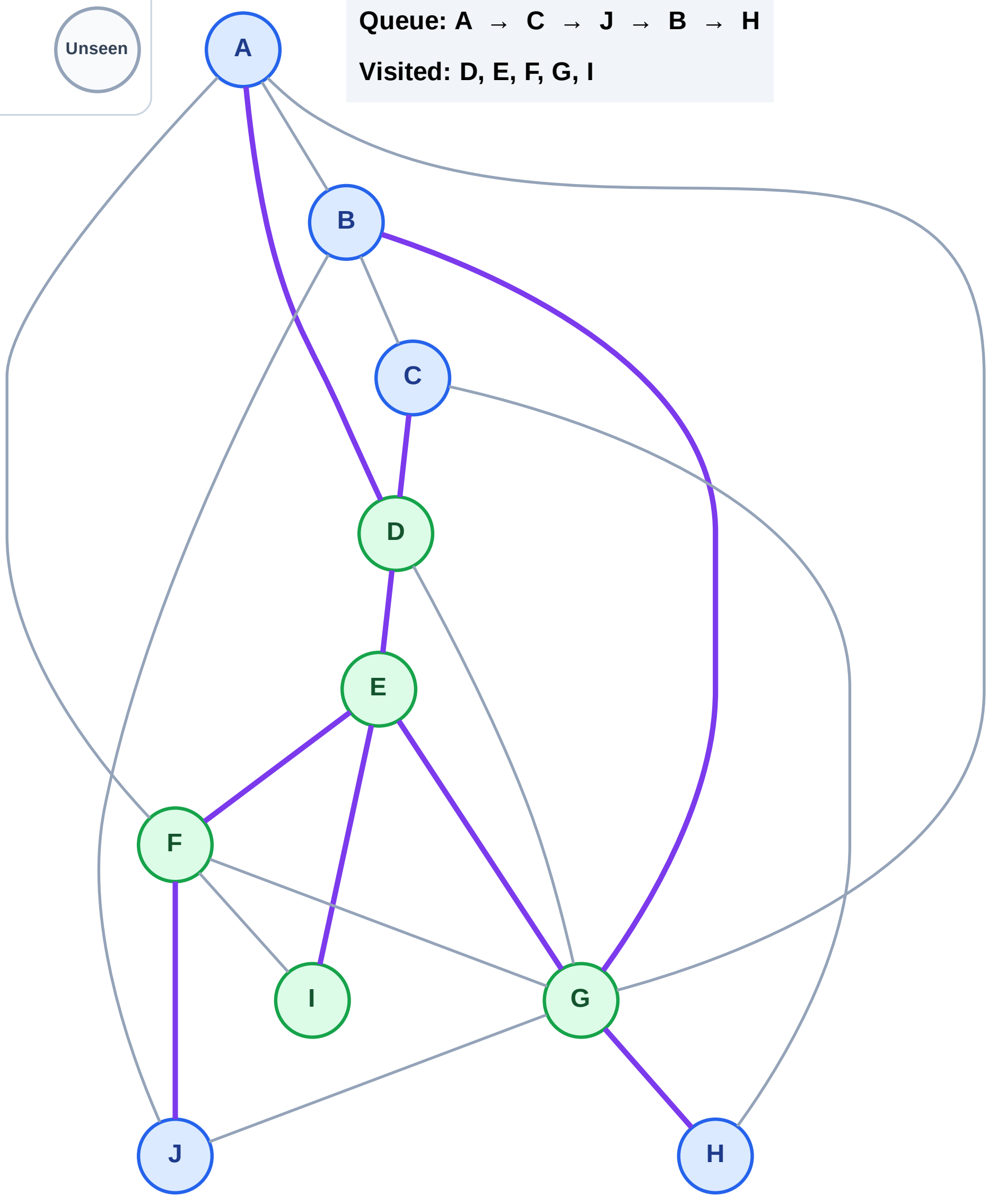
BFS Traversal

Step 11: No new neighbors from I

Legend



Queue: A → C → J → B → H
Visited: D, E, F, G, I



BFS Traversal

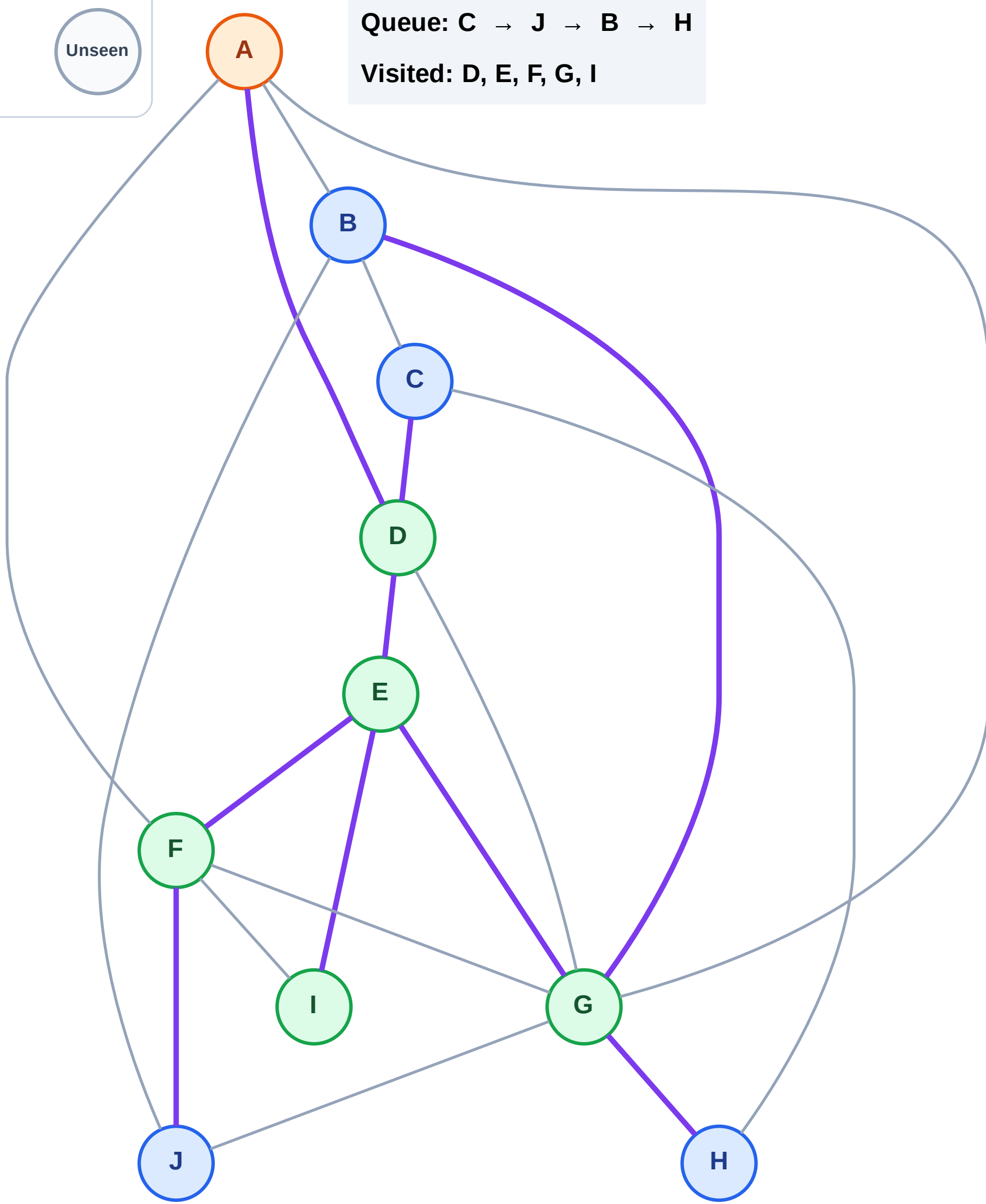
Step 12: Process node A

Legend



Queue: C → J → B → H

Visited: D, E, F, G, I



BFS Traversal

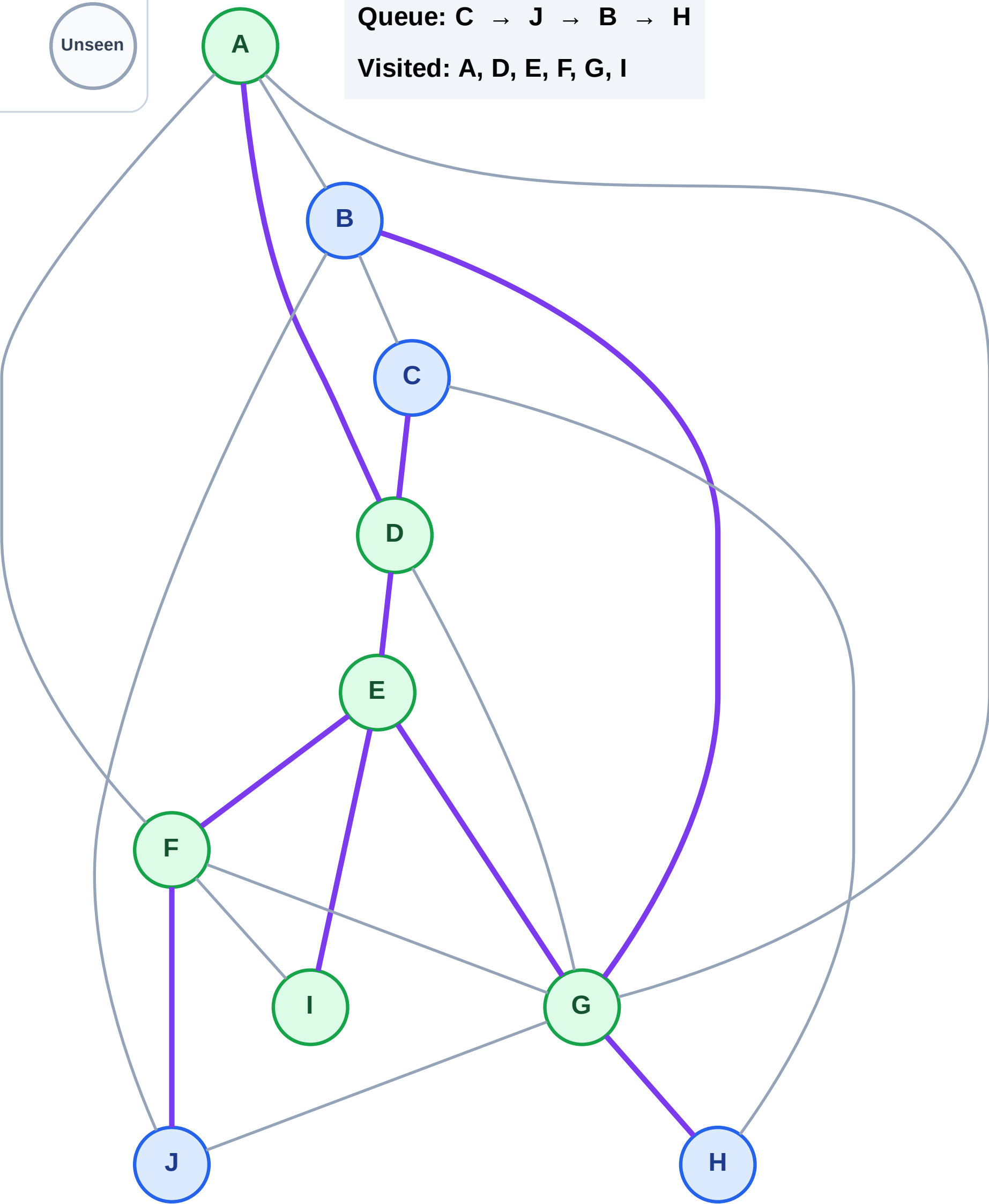
Step 13: No new neighbors from A

Legend



Queue: C → J → B → H

Visited: A, D, E, F, G, I



BFS Traversal

Step 14: Process node C

Legend

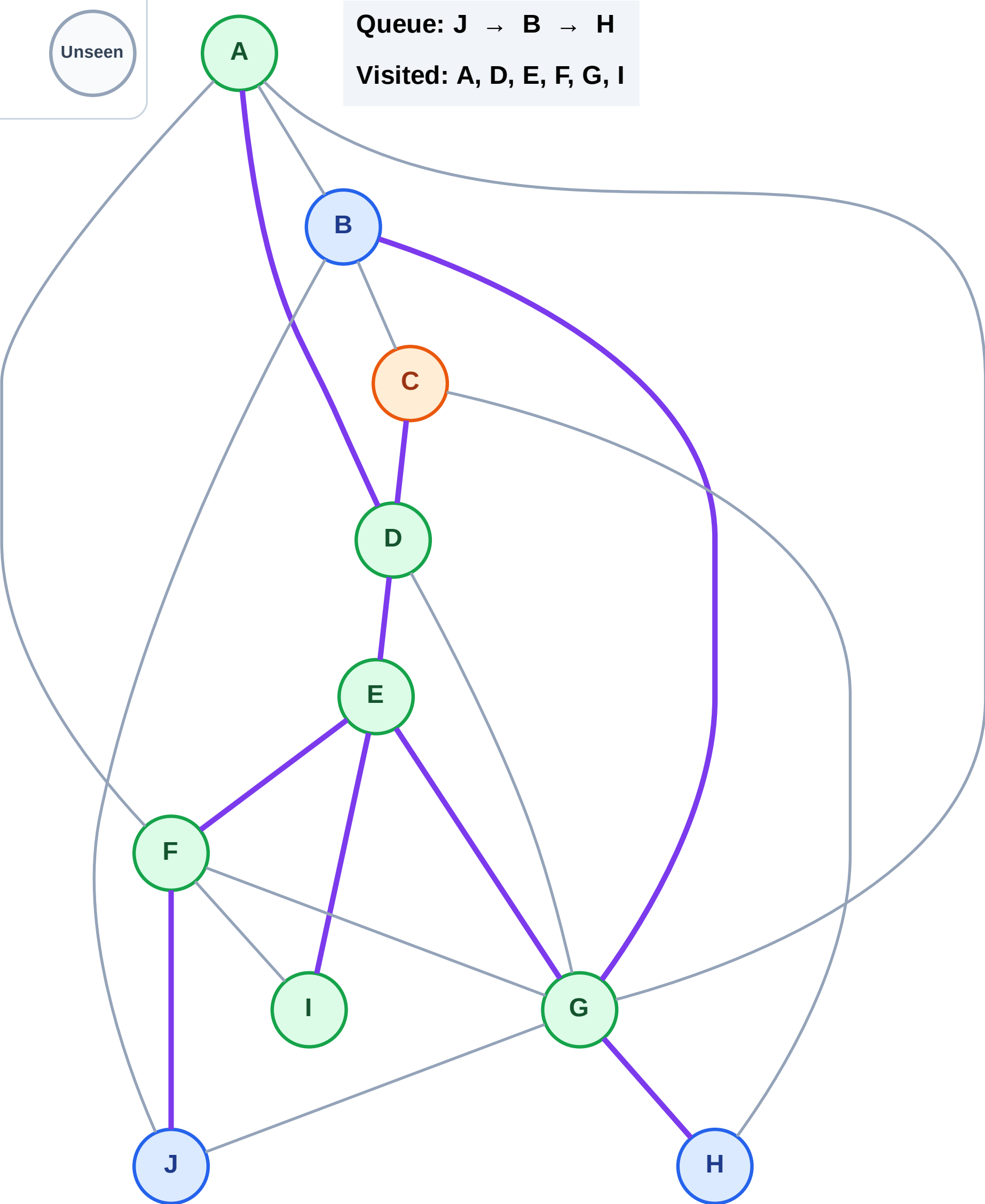
Complete

Processing

Discovered

Unseen

Queue: J → B → H
Visited: A, D, E, F, G, I



BFS Traversal

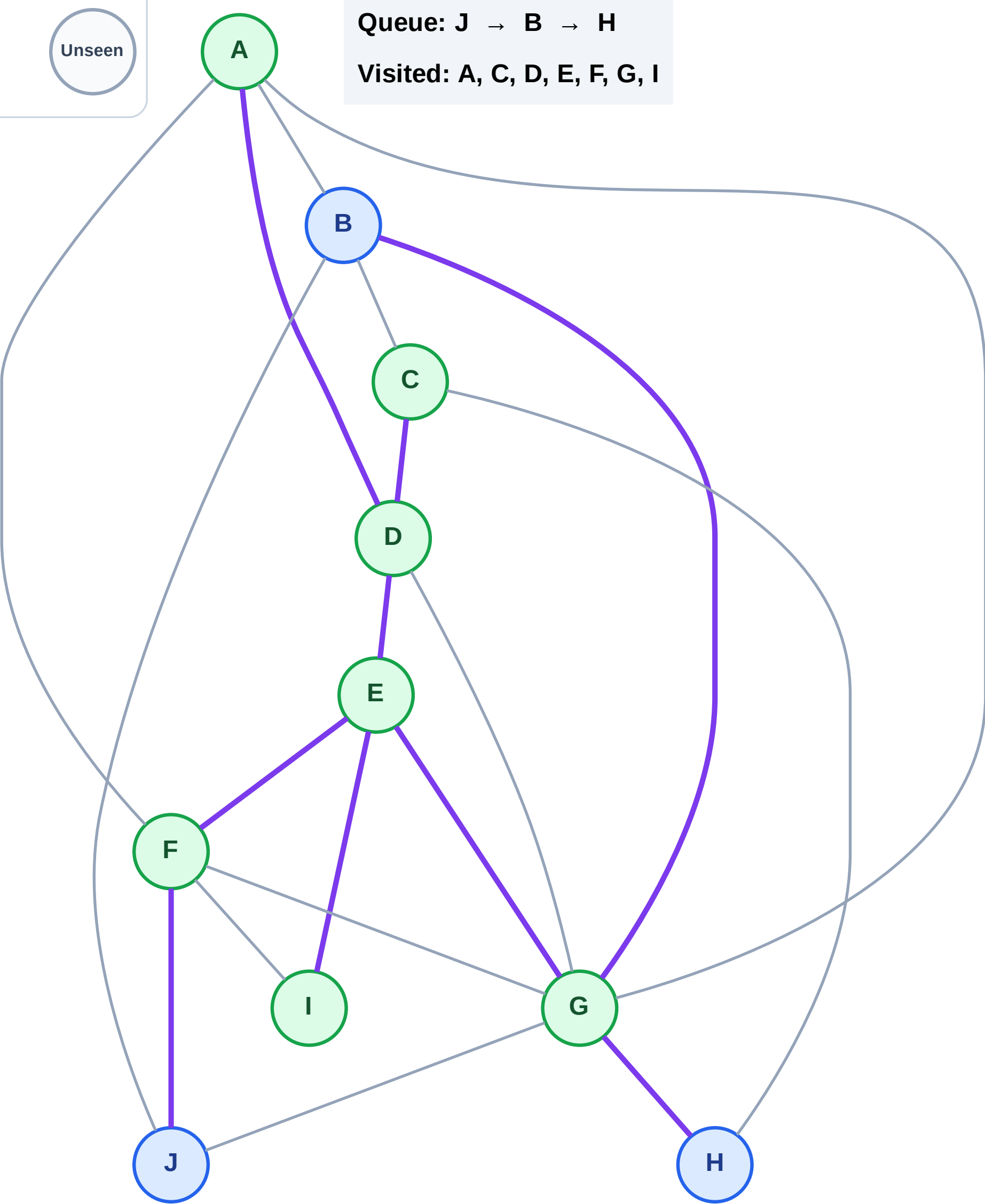
Step 15: No new neighbors from C

Legend



Queue: J → B → H

Visited: A, C, D, E, F, G, I



BFS Traversal

Step 16: Process node J

Legend

Complete

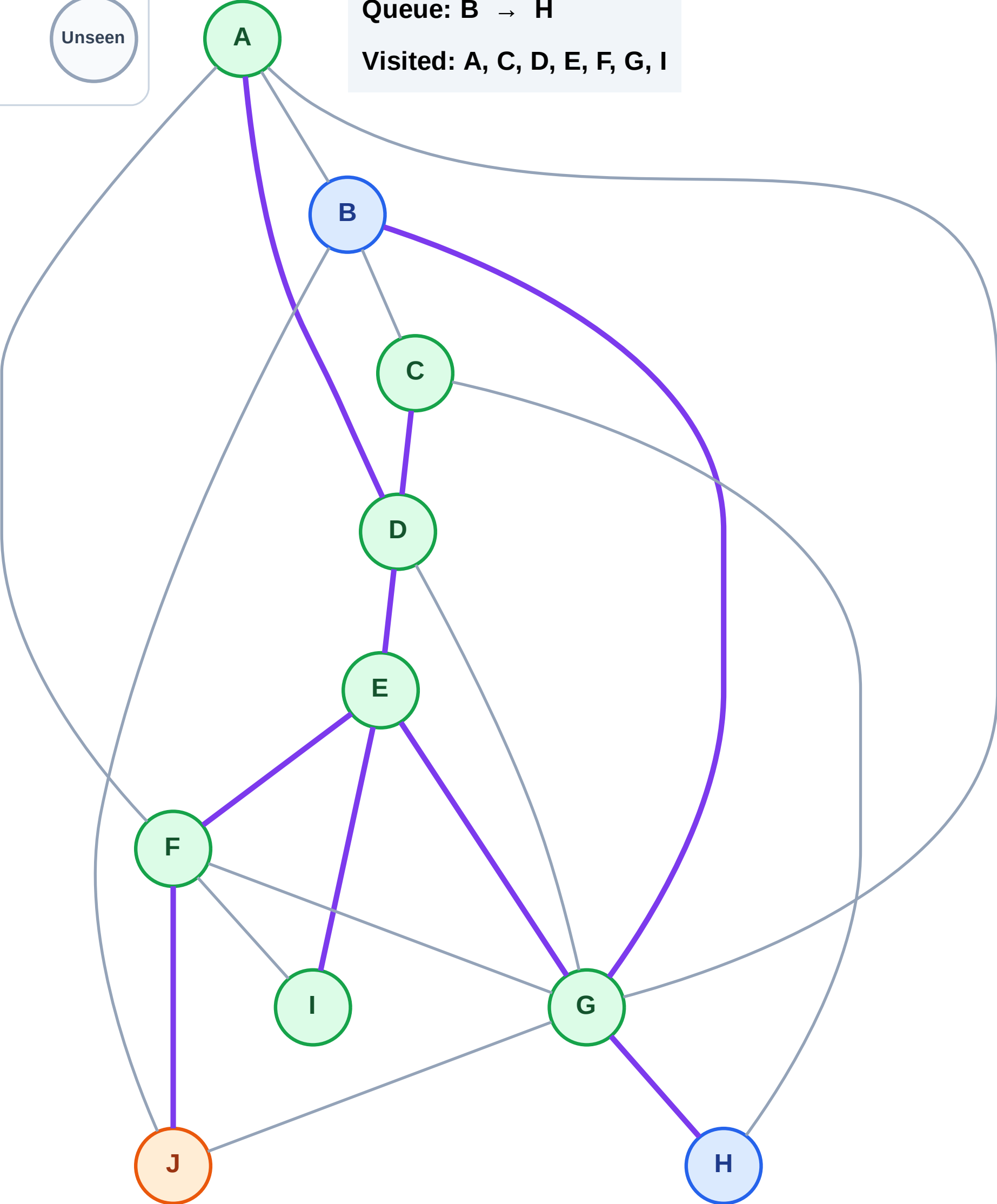
Processing

Discovered

Unseen

Queue: B → H

Visited: A, C, D, E, F, G, I



BFS Traversal

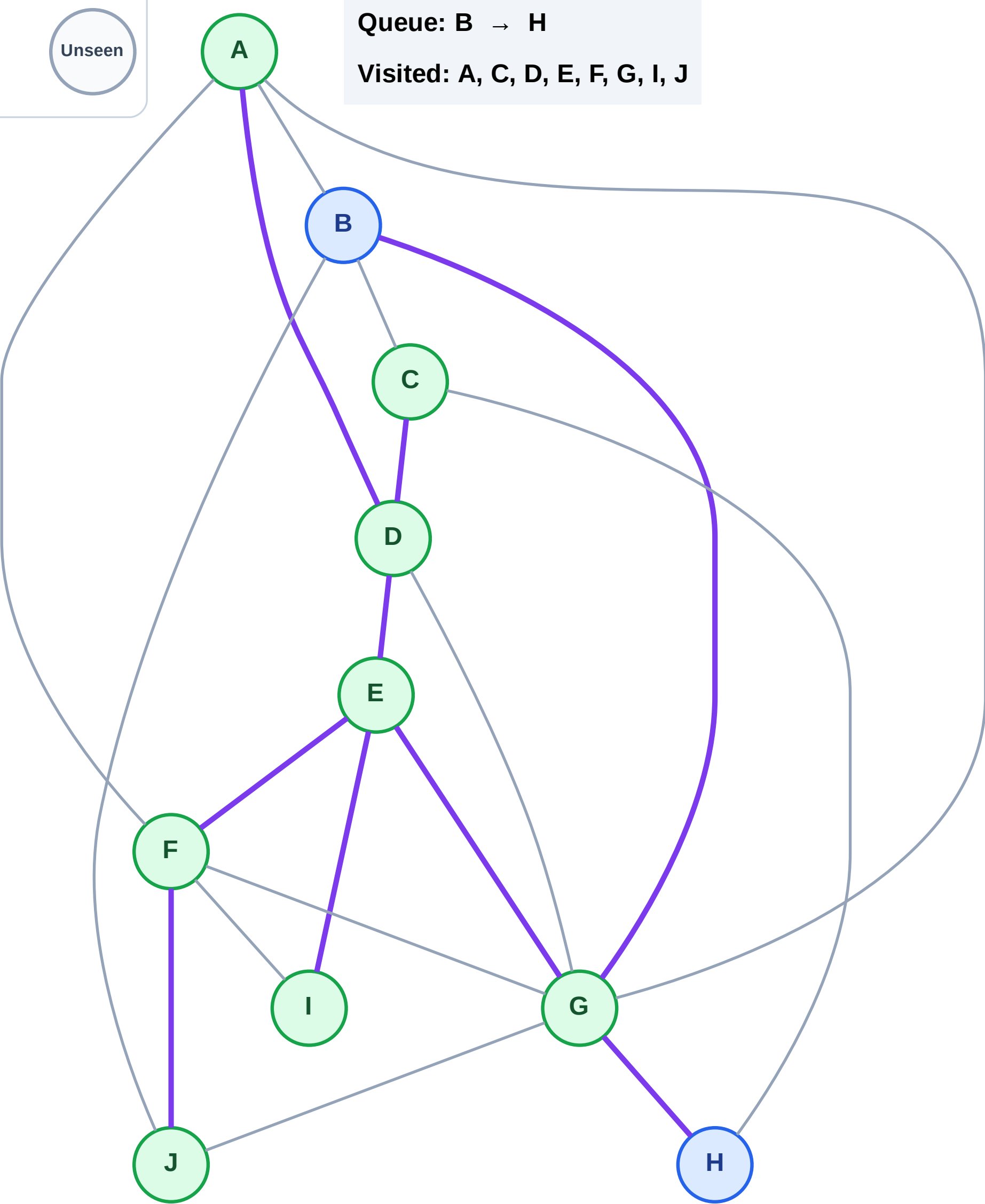
Step 17: No new neighbors from J

Legend



Queue: B → H

Visited: A, C, D, E, F, G, I, J



BFS Traversal

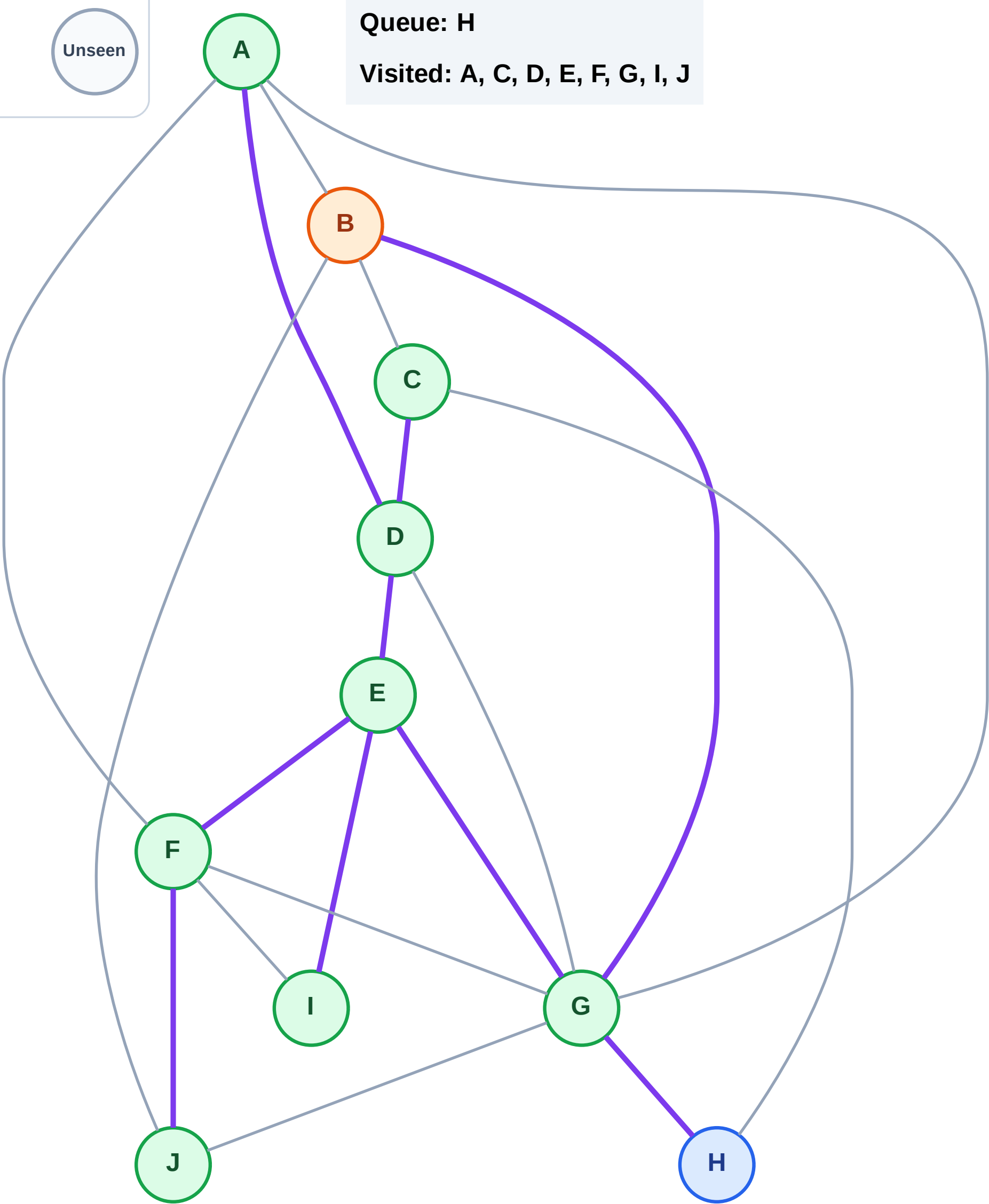
Step 18: Process node B

Legend



Queue: H

Visited: A, C, D, E, F, G, I, J



Step 19: No new neighbors from B

Step 19: No new neighbors from B



BFS Traversal

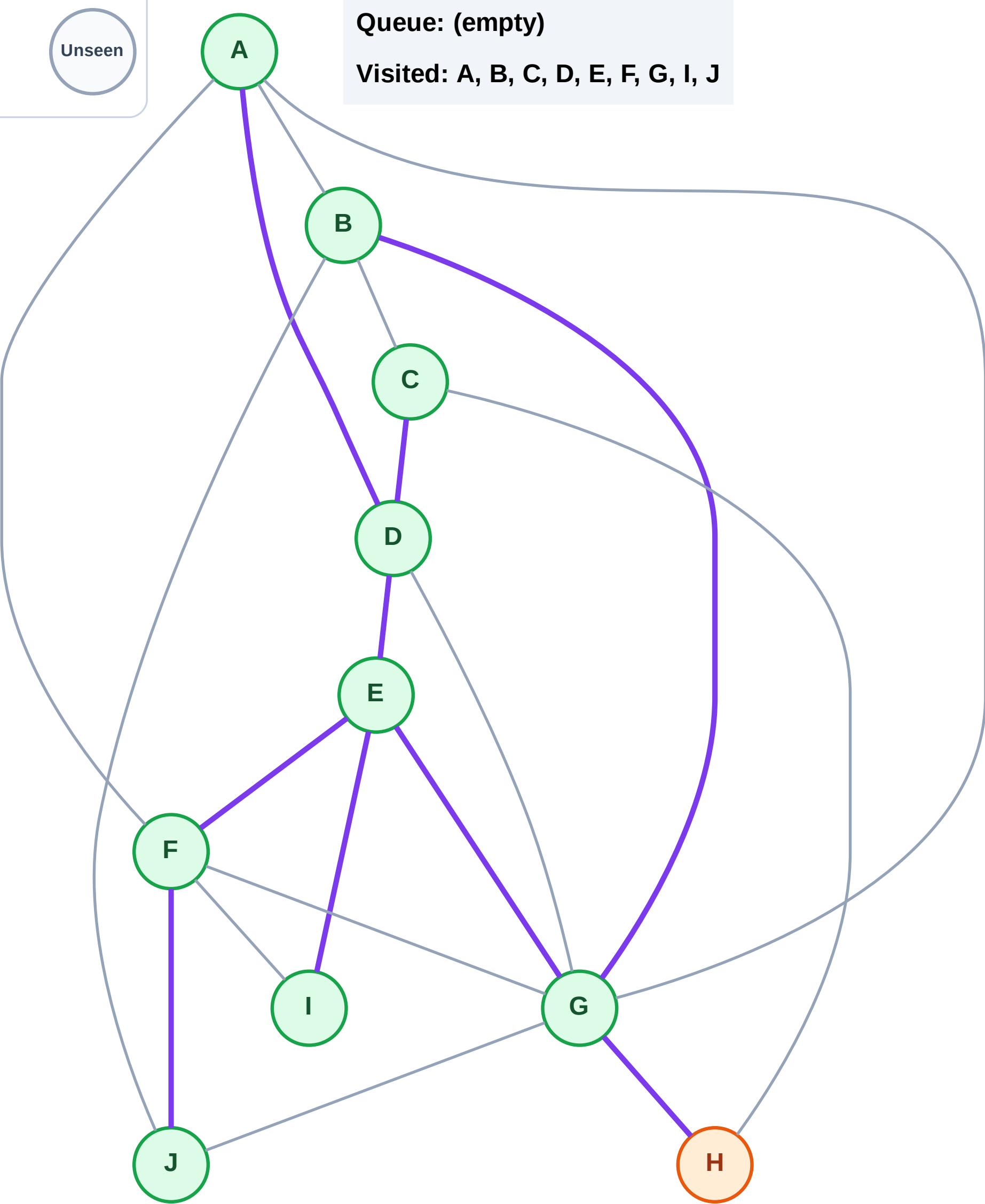
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

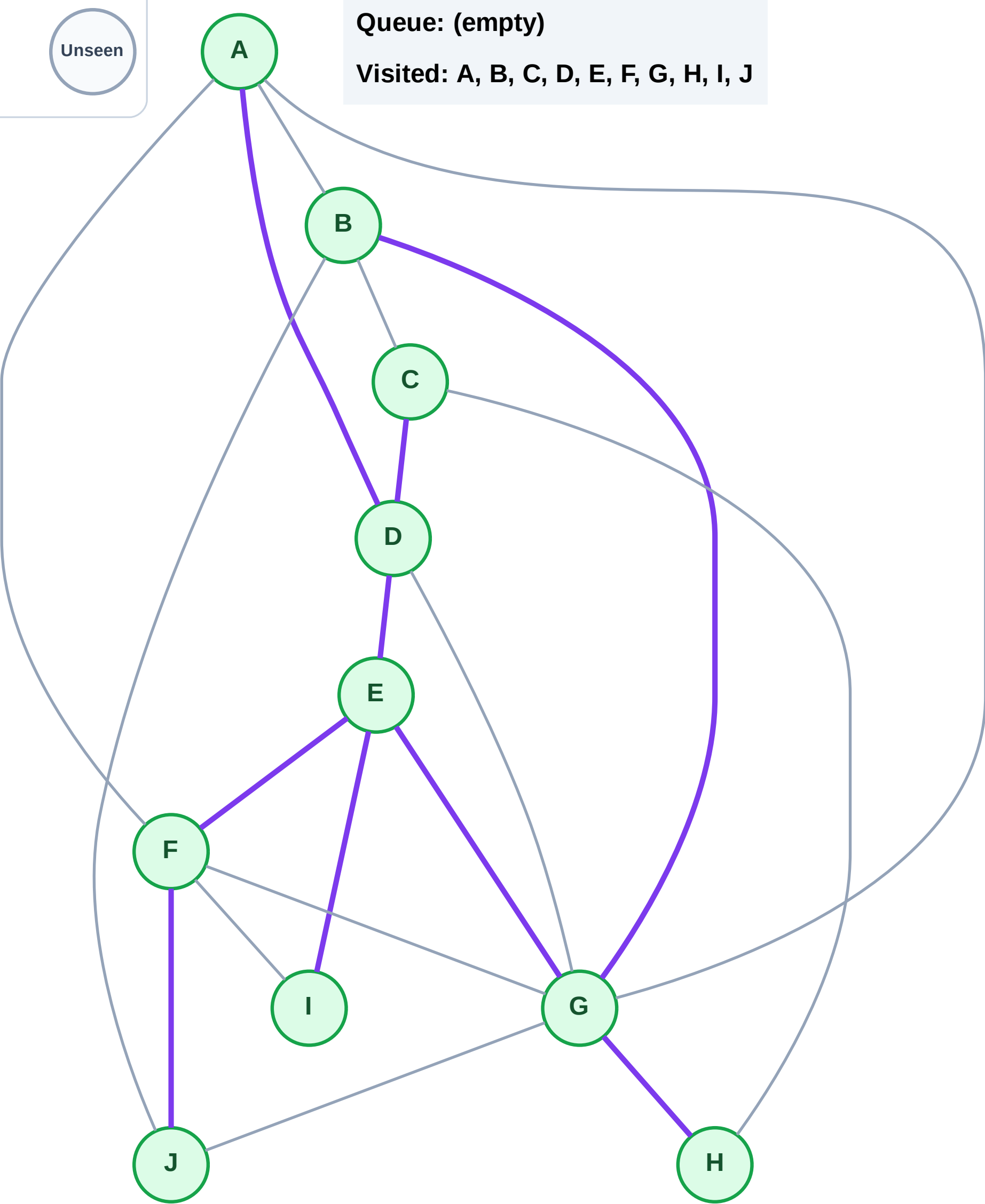
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

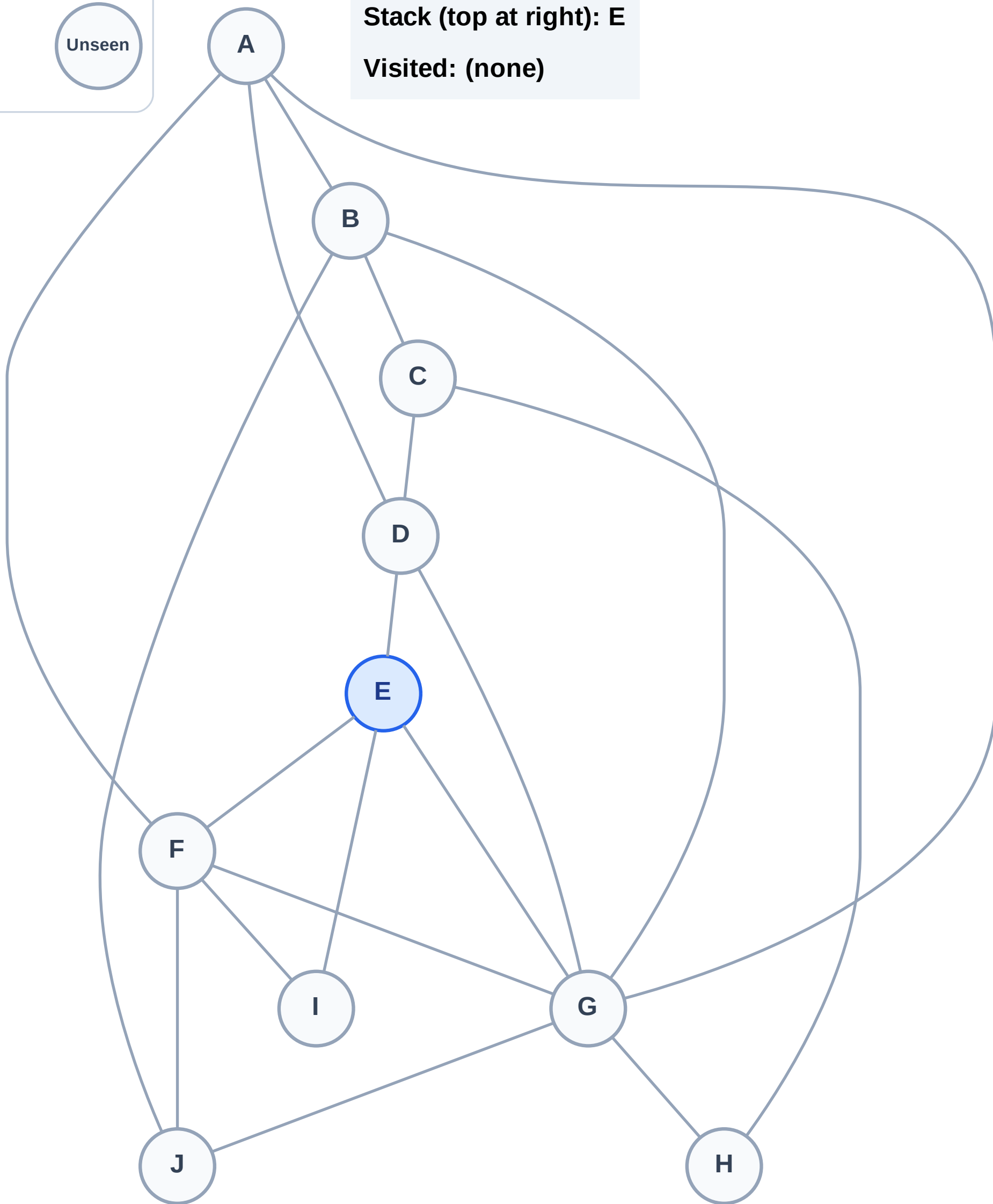
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

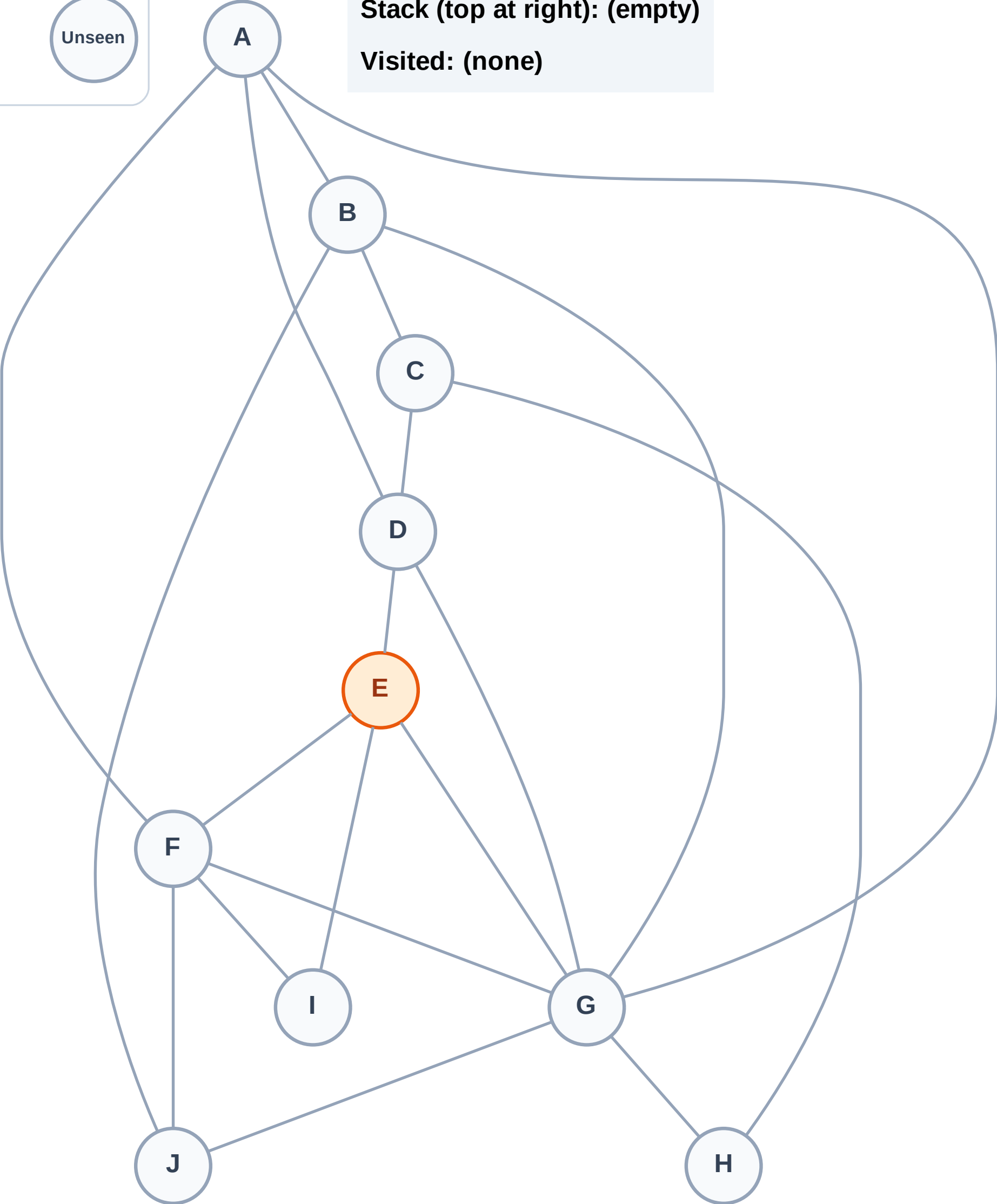
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



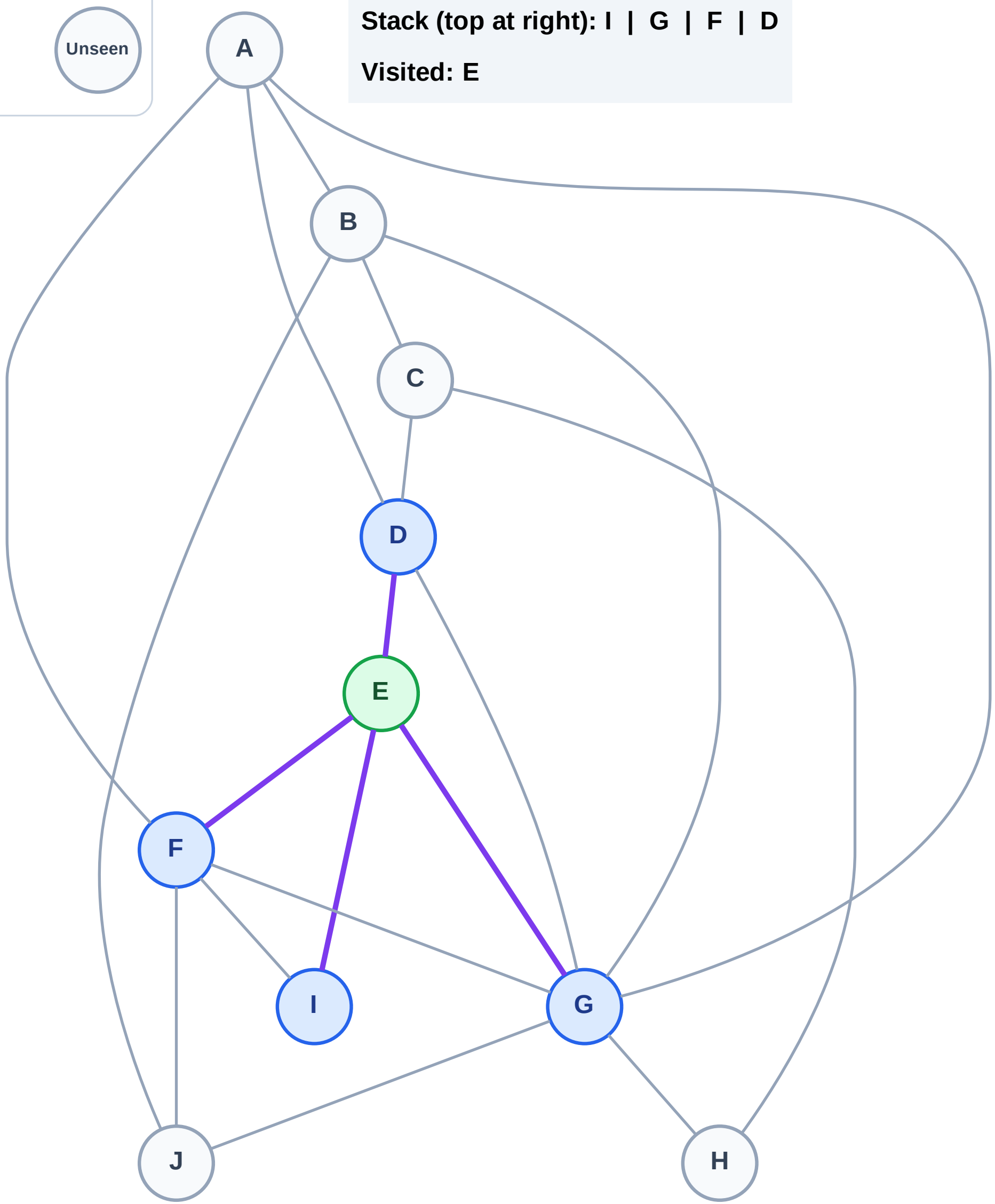
DFS Traversal

Step 3: Discover I, G, F, D from E

Legend



Stack (top at right): I | G | F | D
Visited: E



DFS Traversal

Step 4: Process node D

Legend

Complete

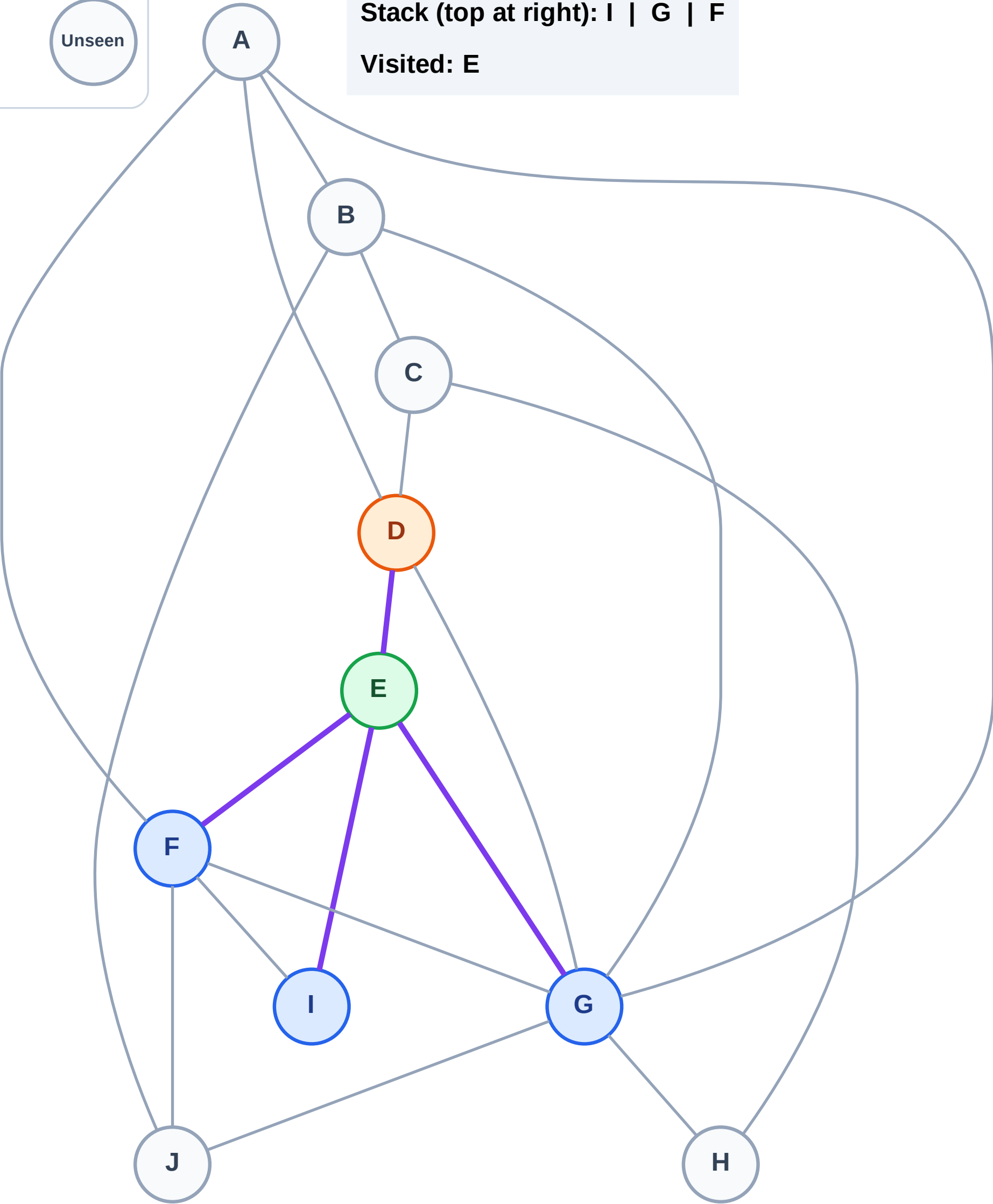
Processing

Discovered

Unseen

Stack (top at right): I | G | F

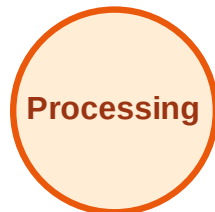
Visited: E



DFS Traversal

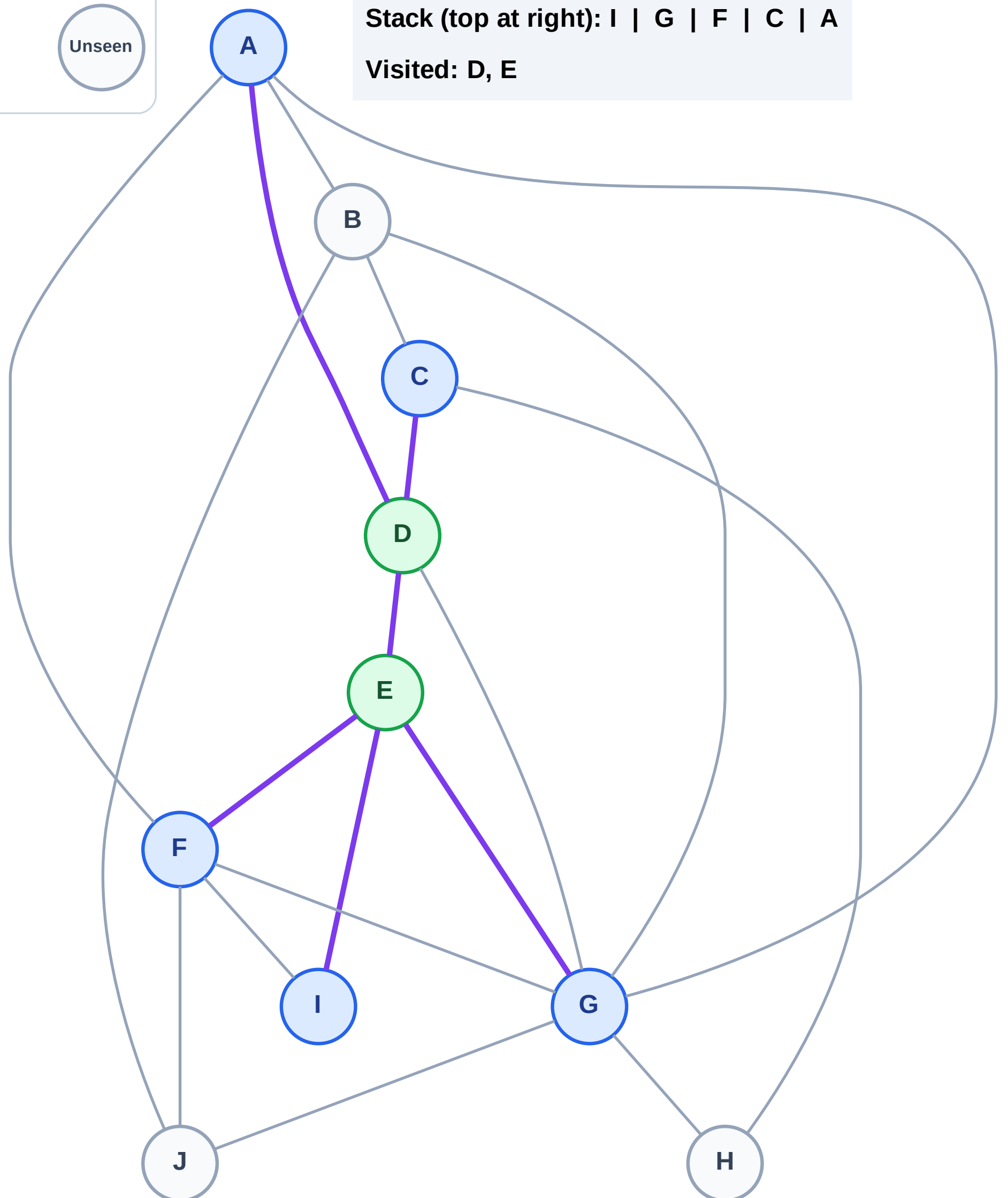
Step 5: Discover C, A from D

Legend



Stack (top at right): I | G | F | C | A

Visited: D, E



DFS Traversal

Step 6: Process node A

Legend

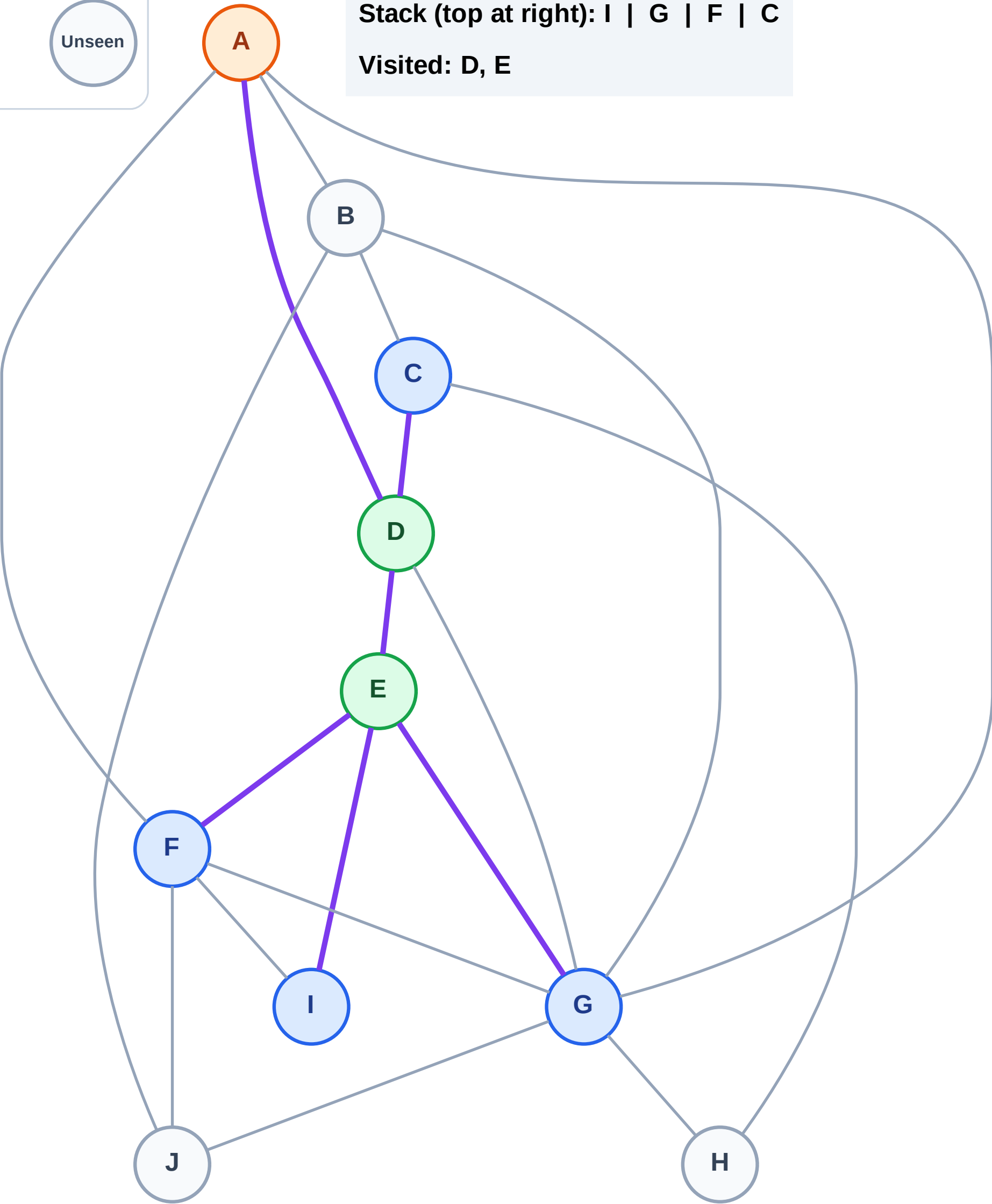
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | F | C
Visited: D, E



DFS Traversal

Step 7: Discover B from A

Legend

Complete

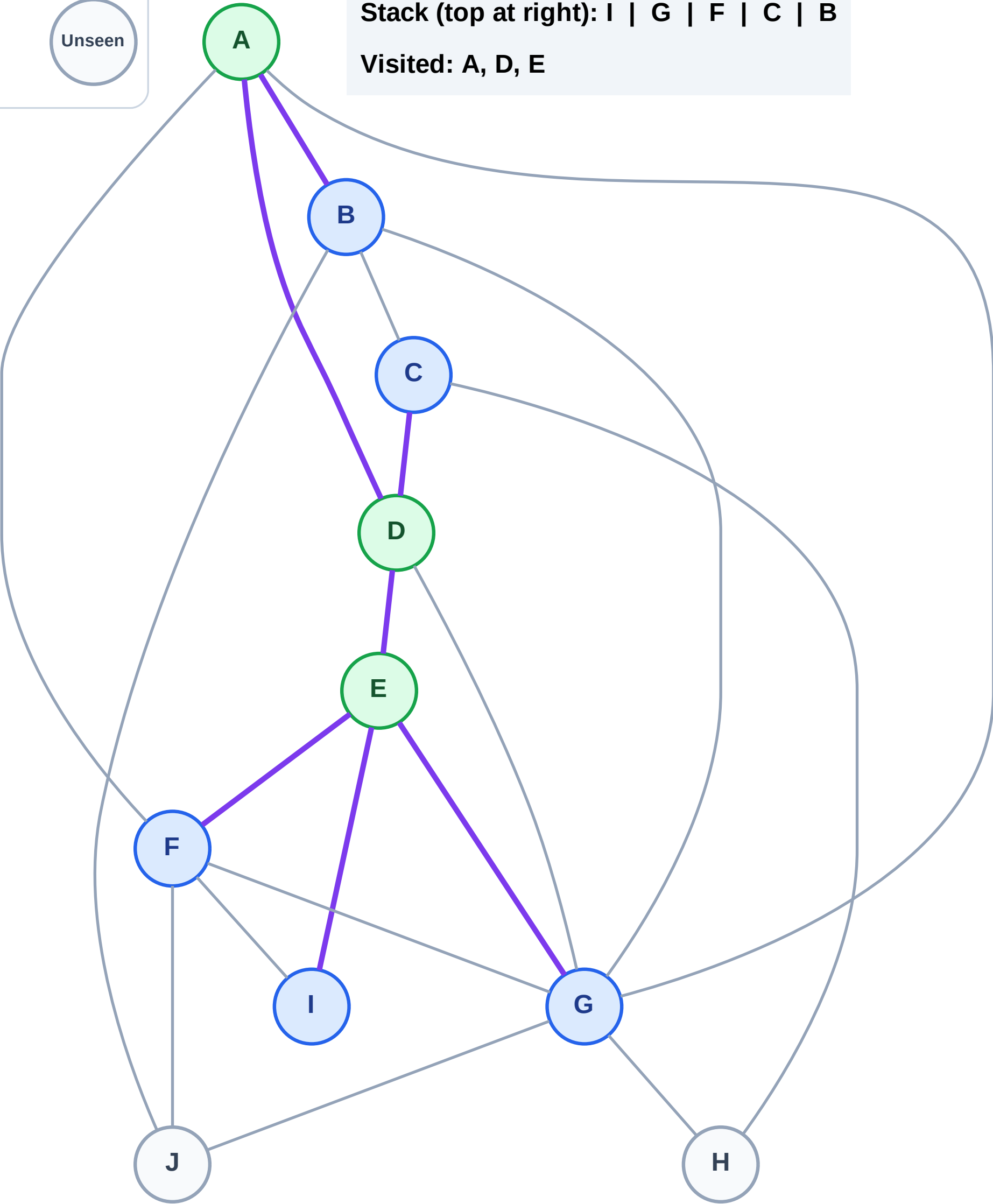
Processing

Discovered

Unseen

Stack (top at right): I | G | F | C | B

Visited: A, D, E



DFS Traversal

Step 8: Process node B

Legend

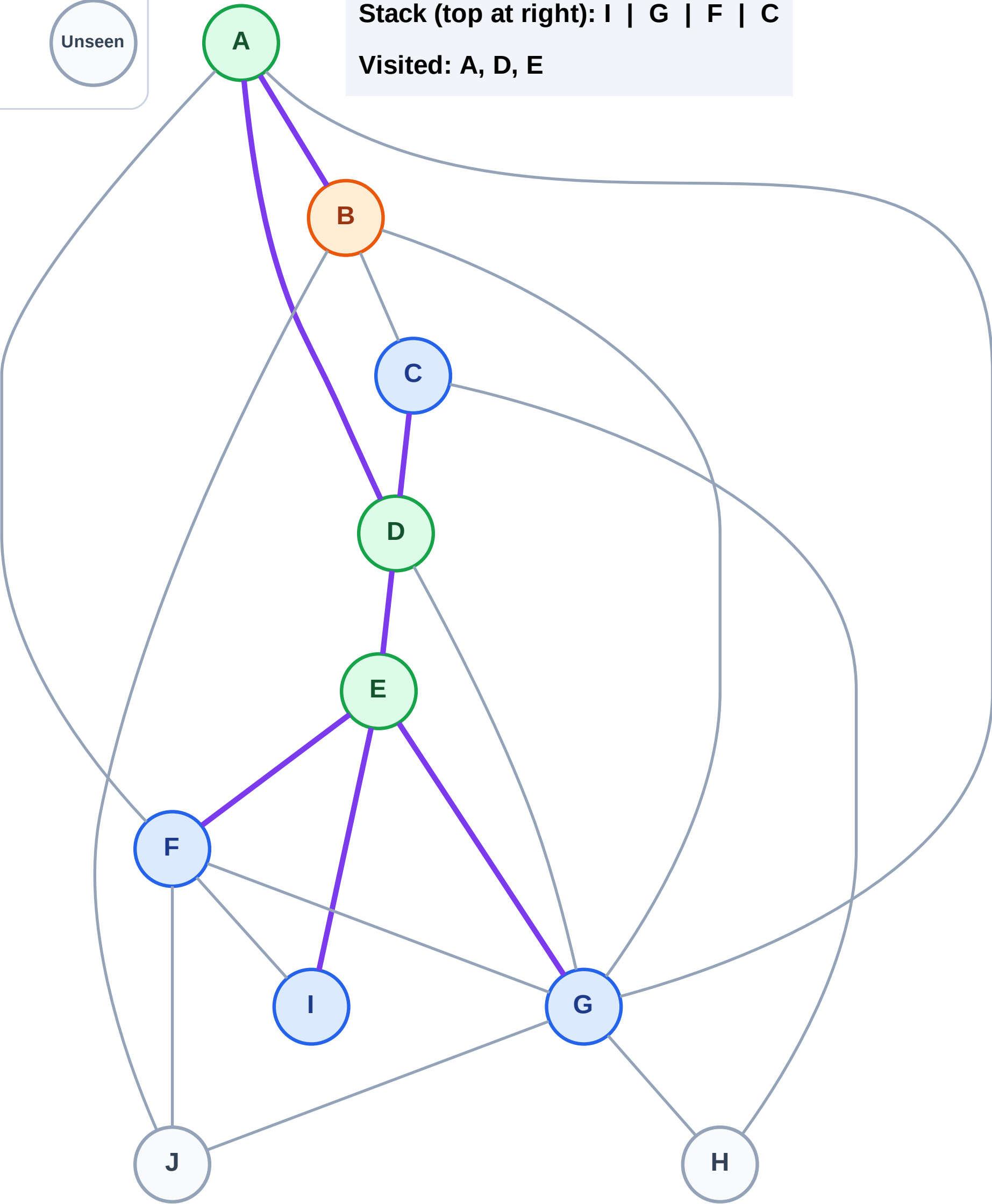
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | F | C
Visited: A, D, E



DFS Traversal

Step 9: Discover J from B

Legend

Complete

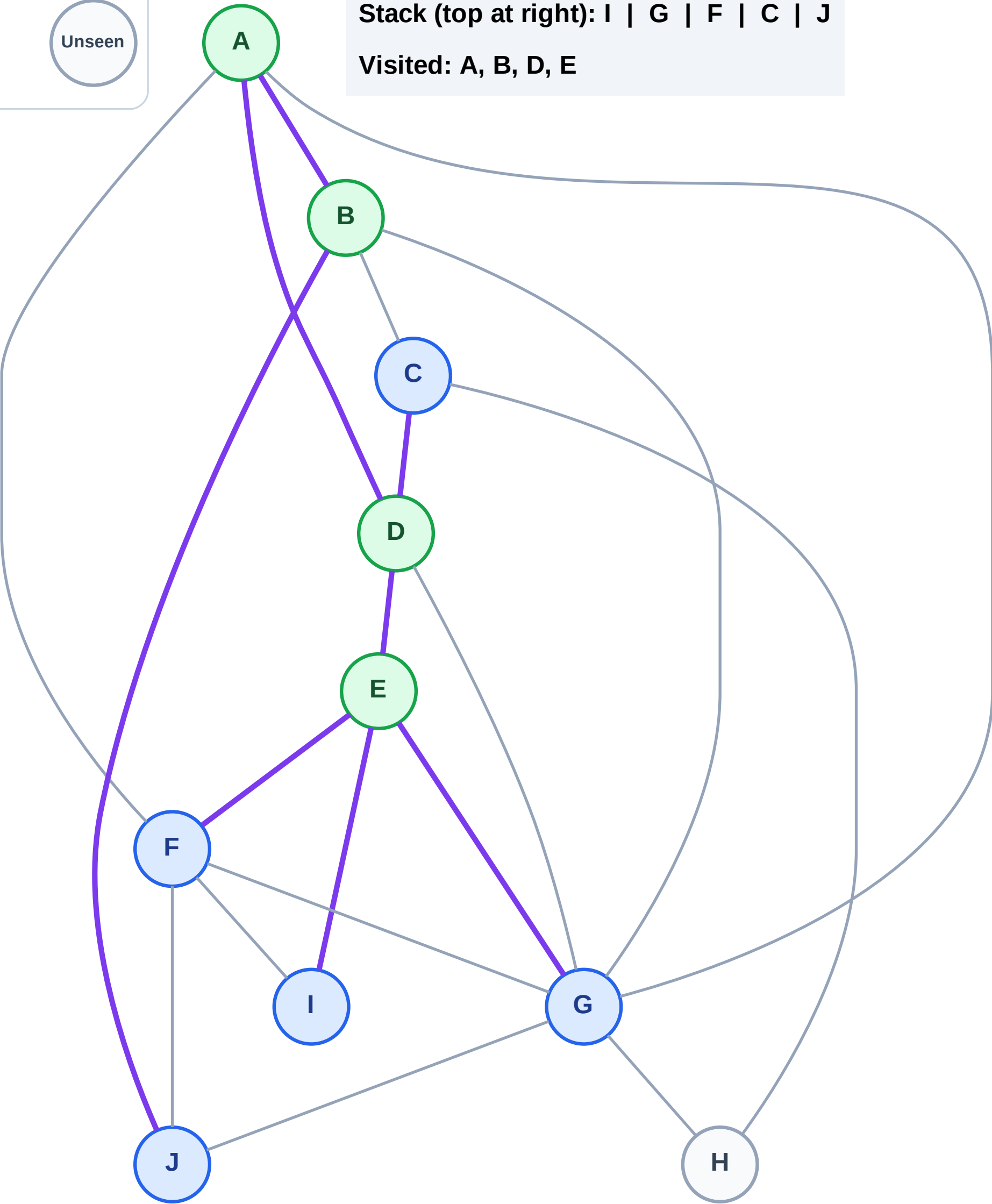
Processing

Discovered

Unseen

Stack (top at right): I | G | F | C | J

Visited: A, B, D, E



DFS Traversal

Step 10: Process node J

Legend

Complete

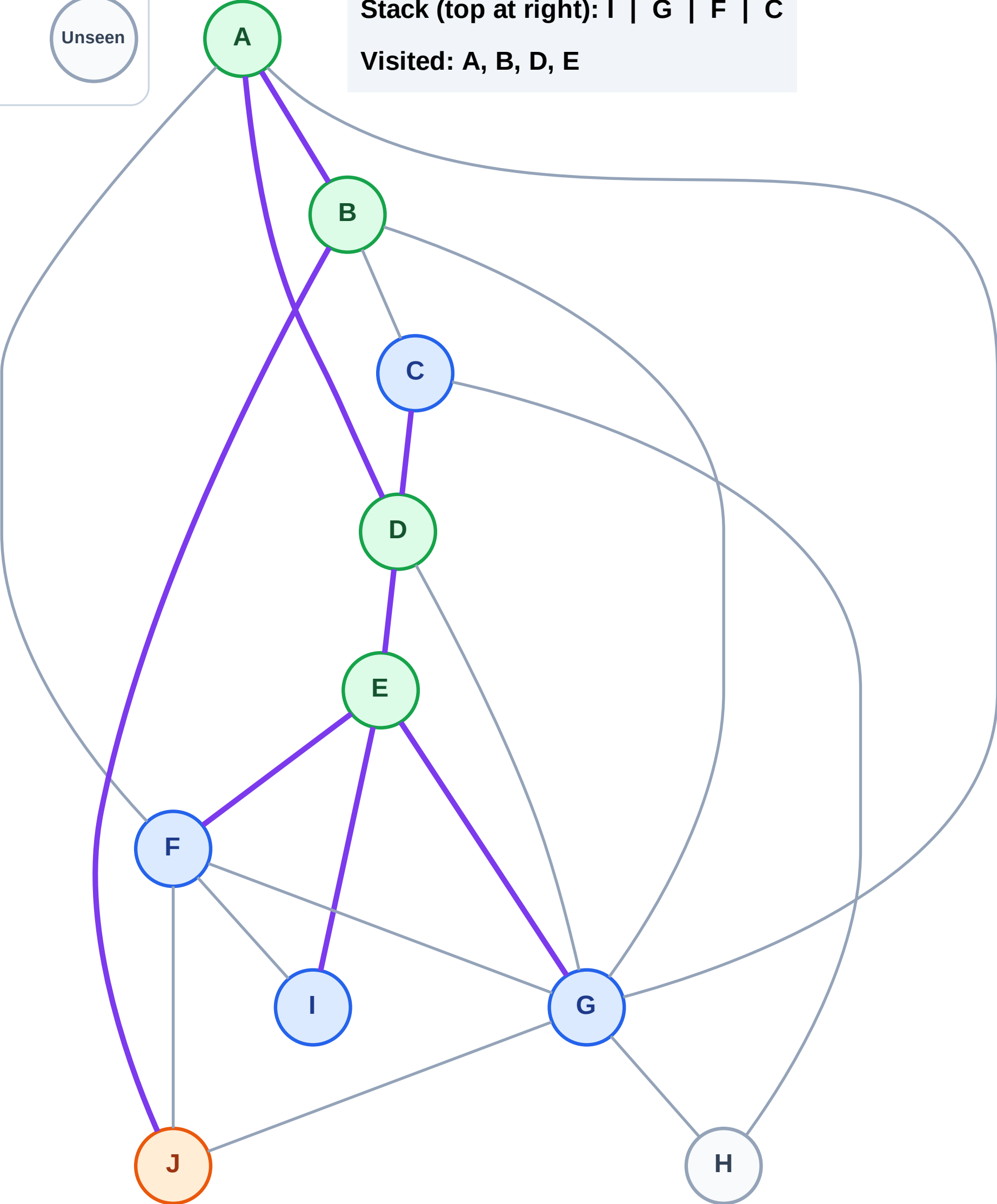
Processing

Discovered

Unseen

Stack (top at right): I | G | F | C

Visited: A, B, D, E



DFS Traversal

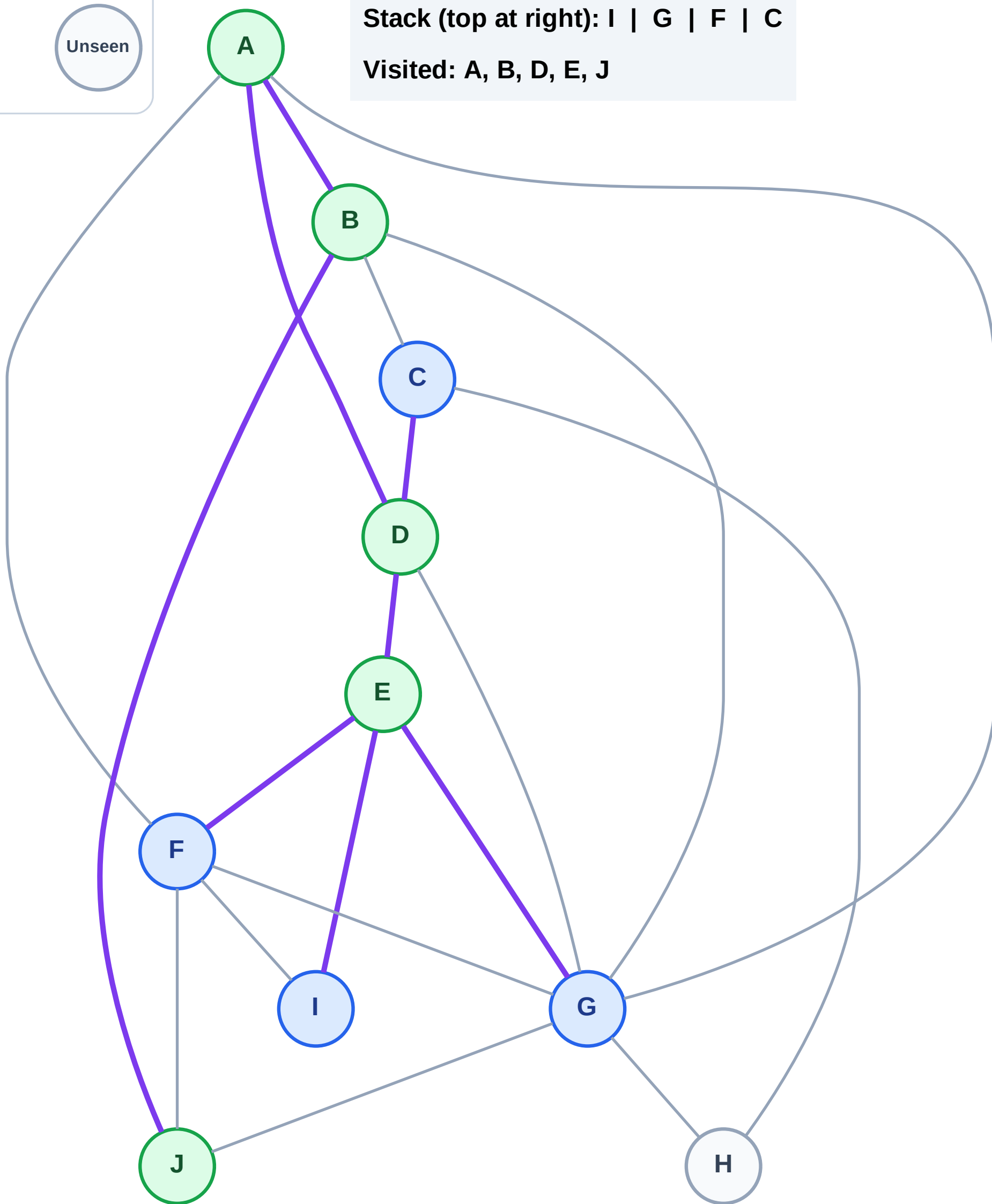
Step 11: No new neighbors from J

Legend



Stack (top at right): I | G | F | C

Visited: A, B, D, E, J



DFS Traversal

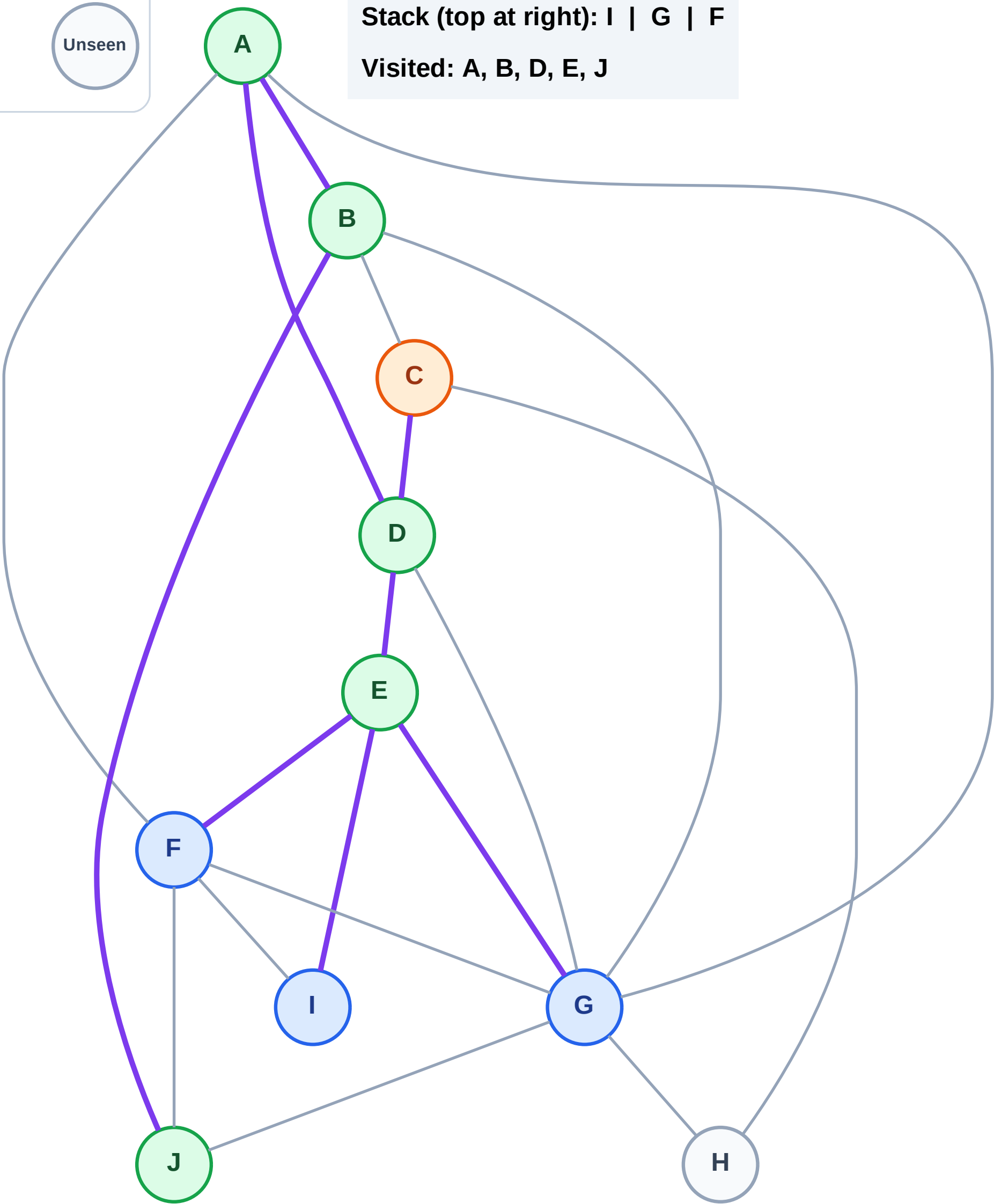
Step 12: Process node C

Legend



Stack (top at right): I | G | F

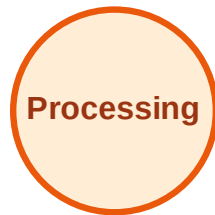
Visited: A, B, D, E, J



DFS Traversal

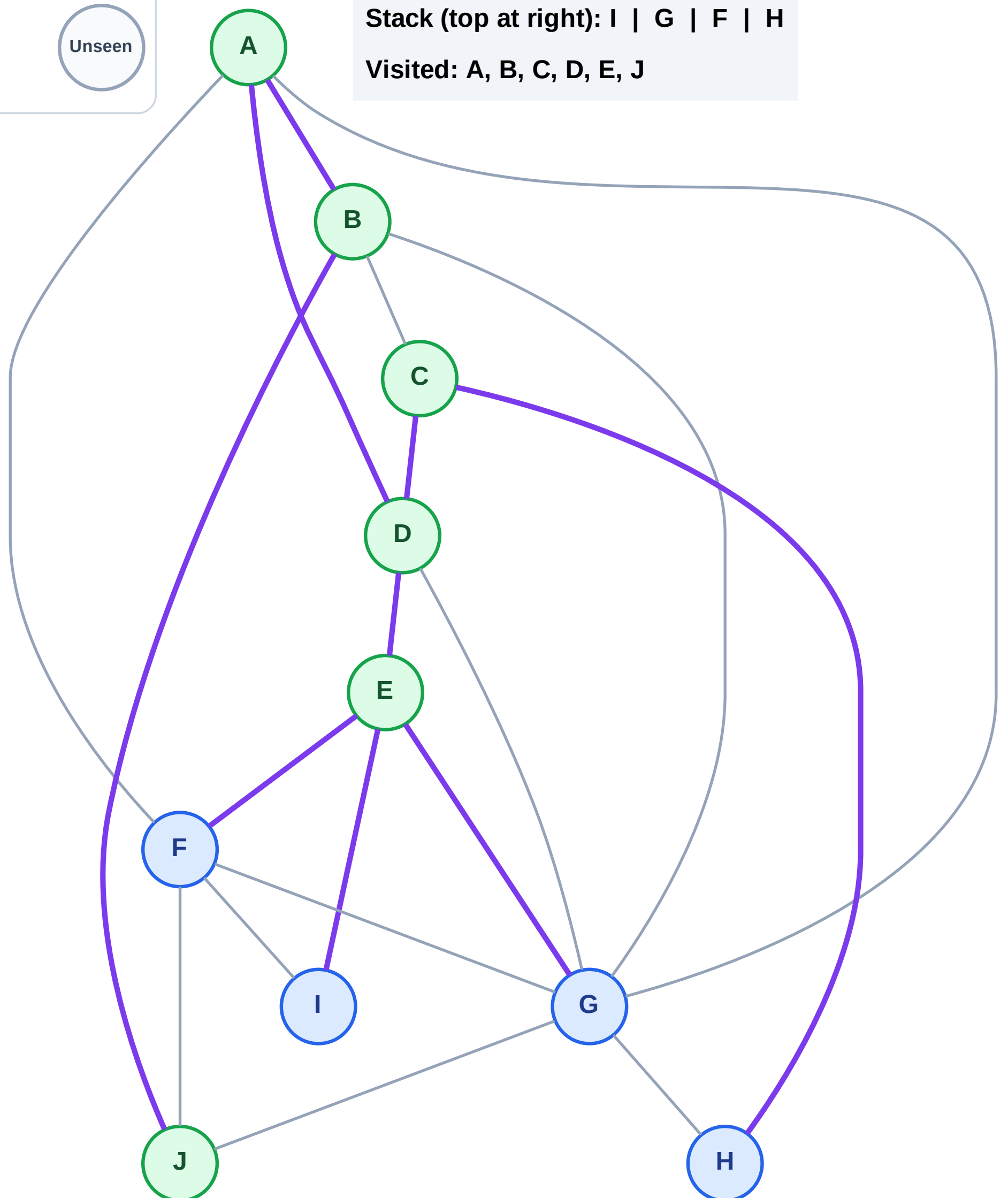
Step 13: Discover H from C

Legend



Stack (top at right): I | G | F | H

Visited: A, B, C, D, E, J



DFS Traversal

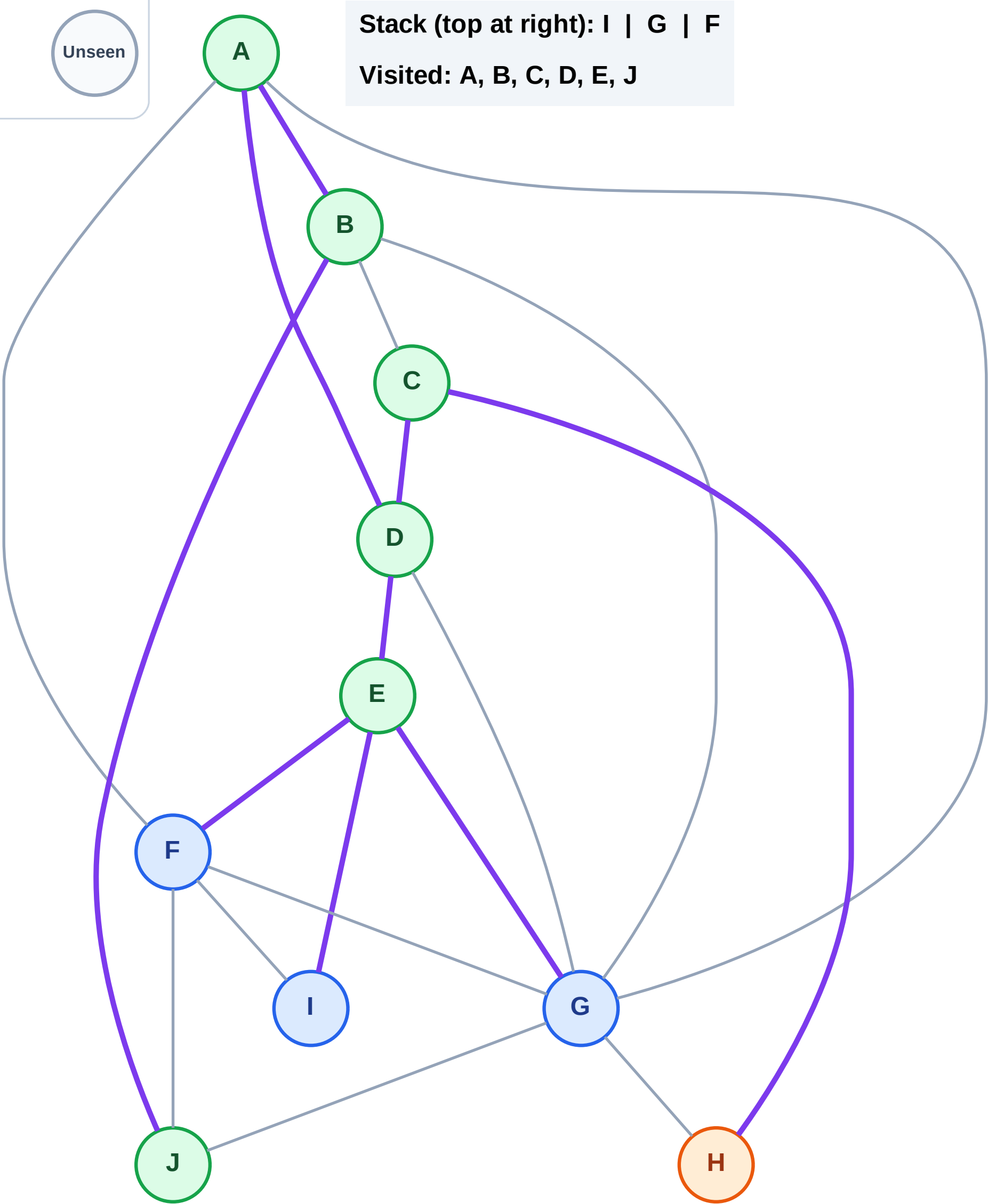
Step 14: Process node H

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, J



DFS Traversal

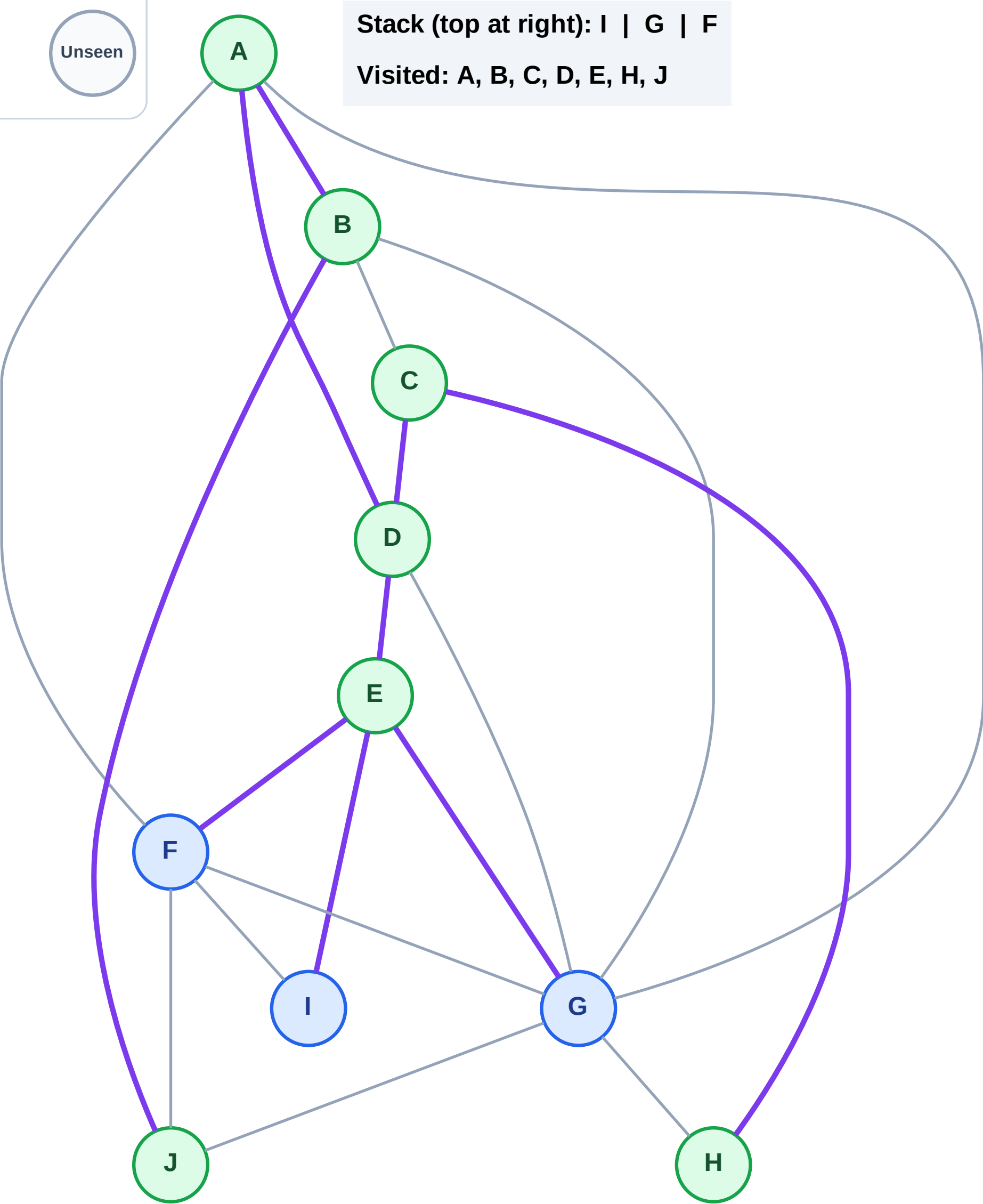
Step 15: No new neighbors from H

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



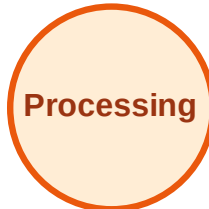
DFS Traversal

Step 16: Process node F

Legend



Complete



Processing



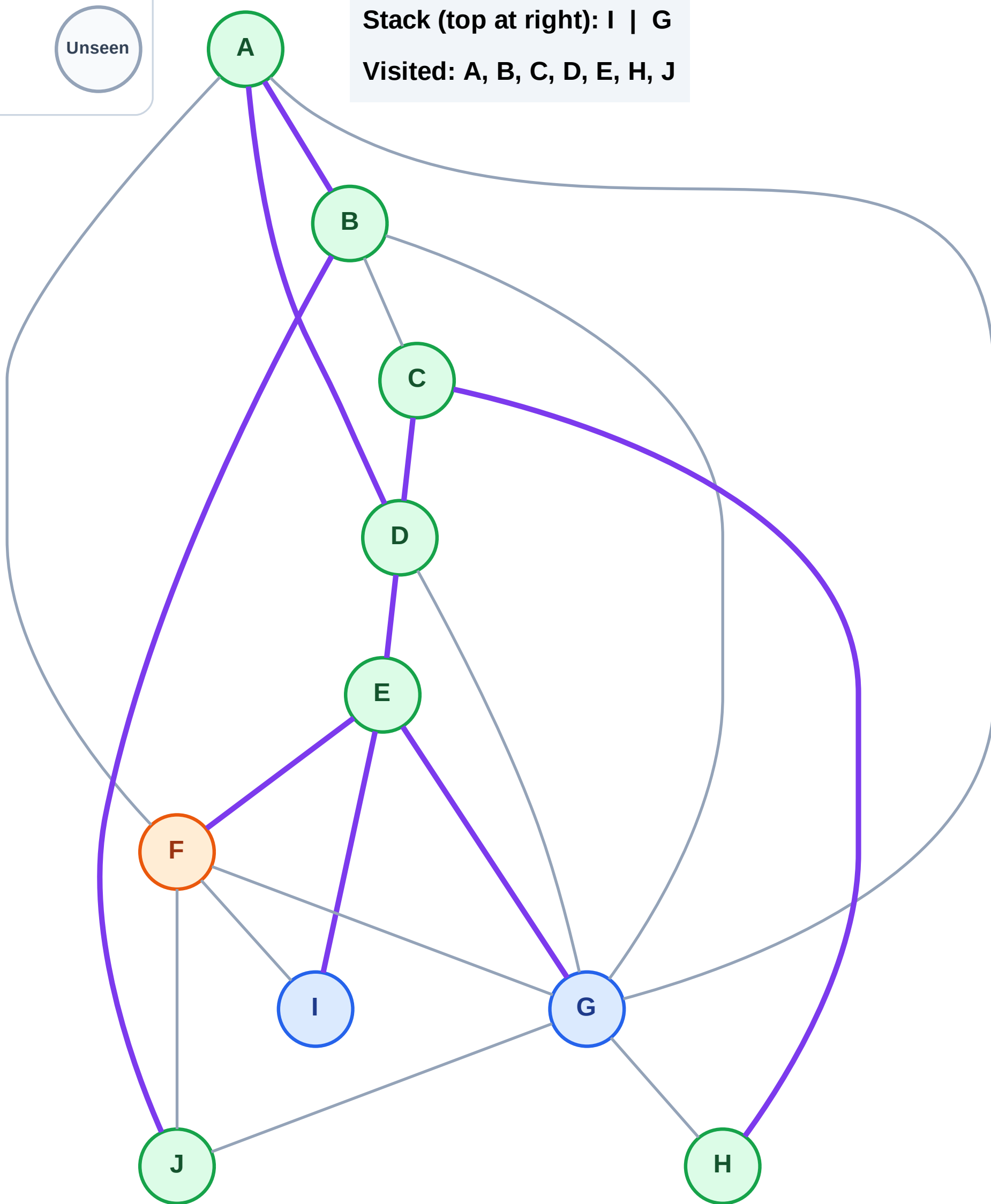
Discovered



Unseen

Stack (top at right): I | G

Visited: A, B, C, D, E, H, J



DFS Traversal

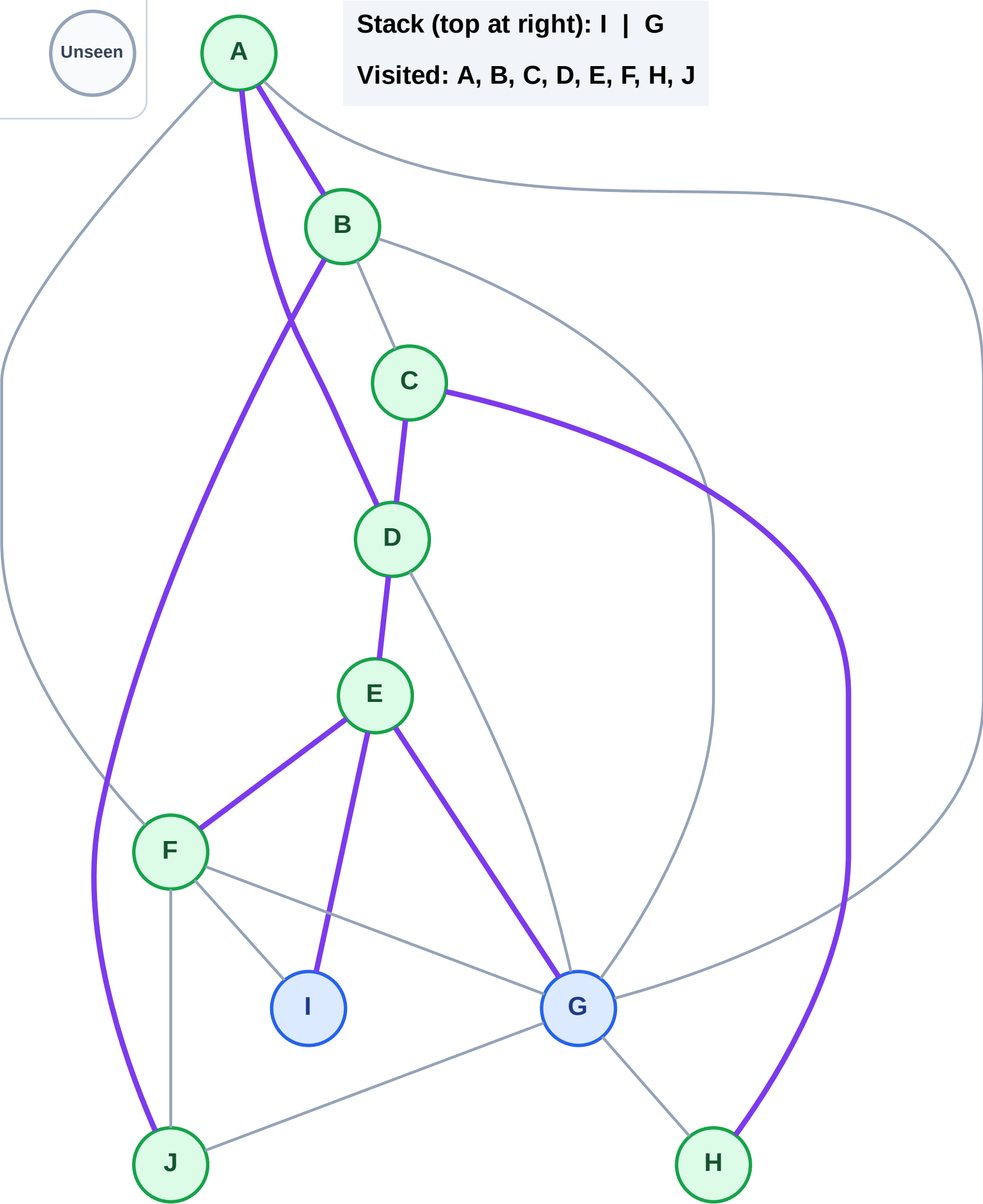
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

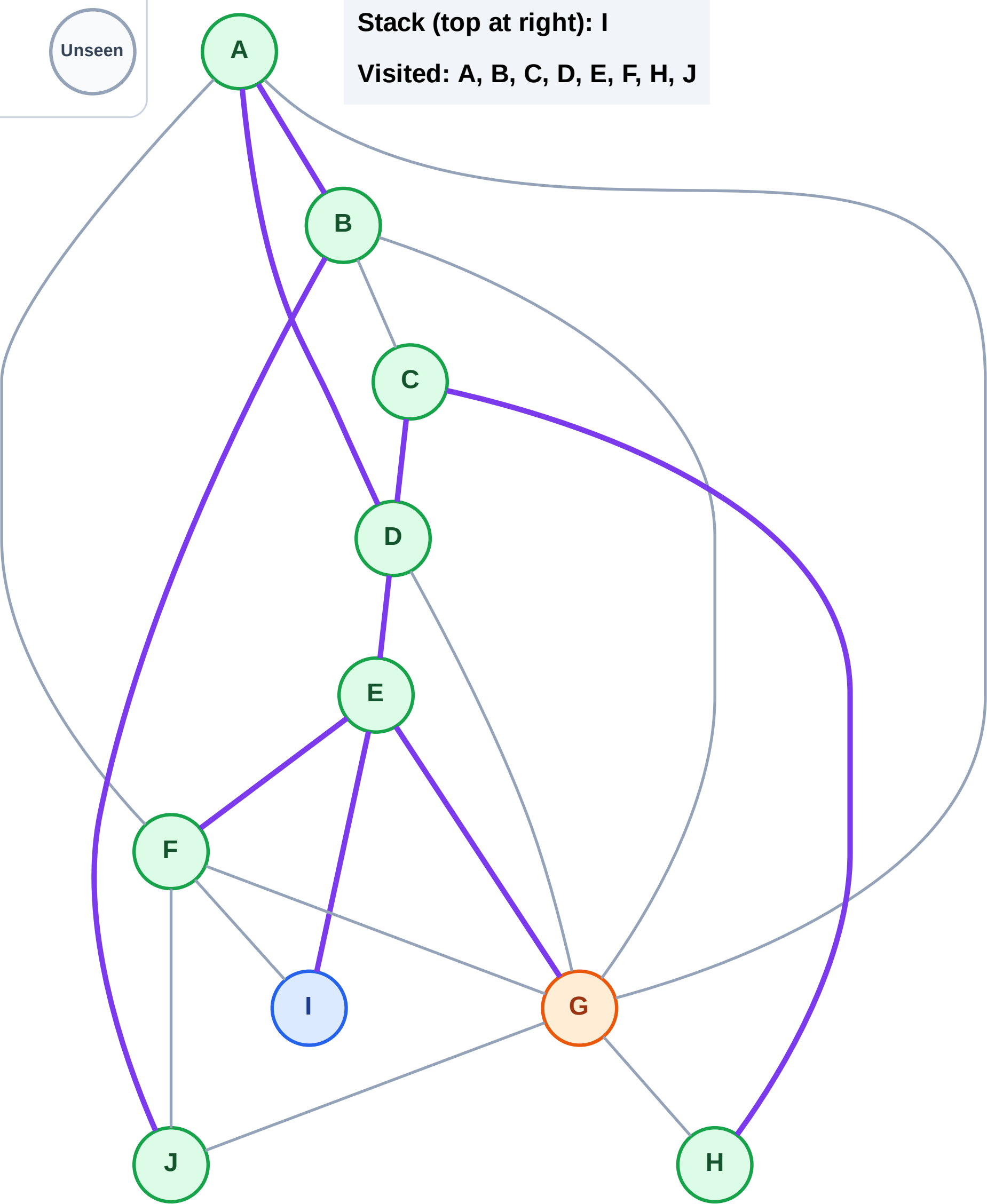
Step 18: Process node G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, H, J



DFS Traversal

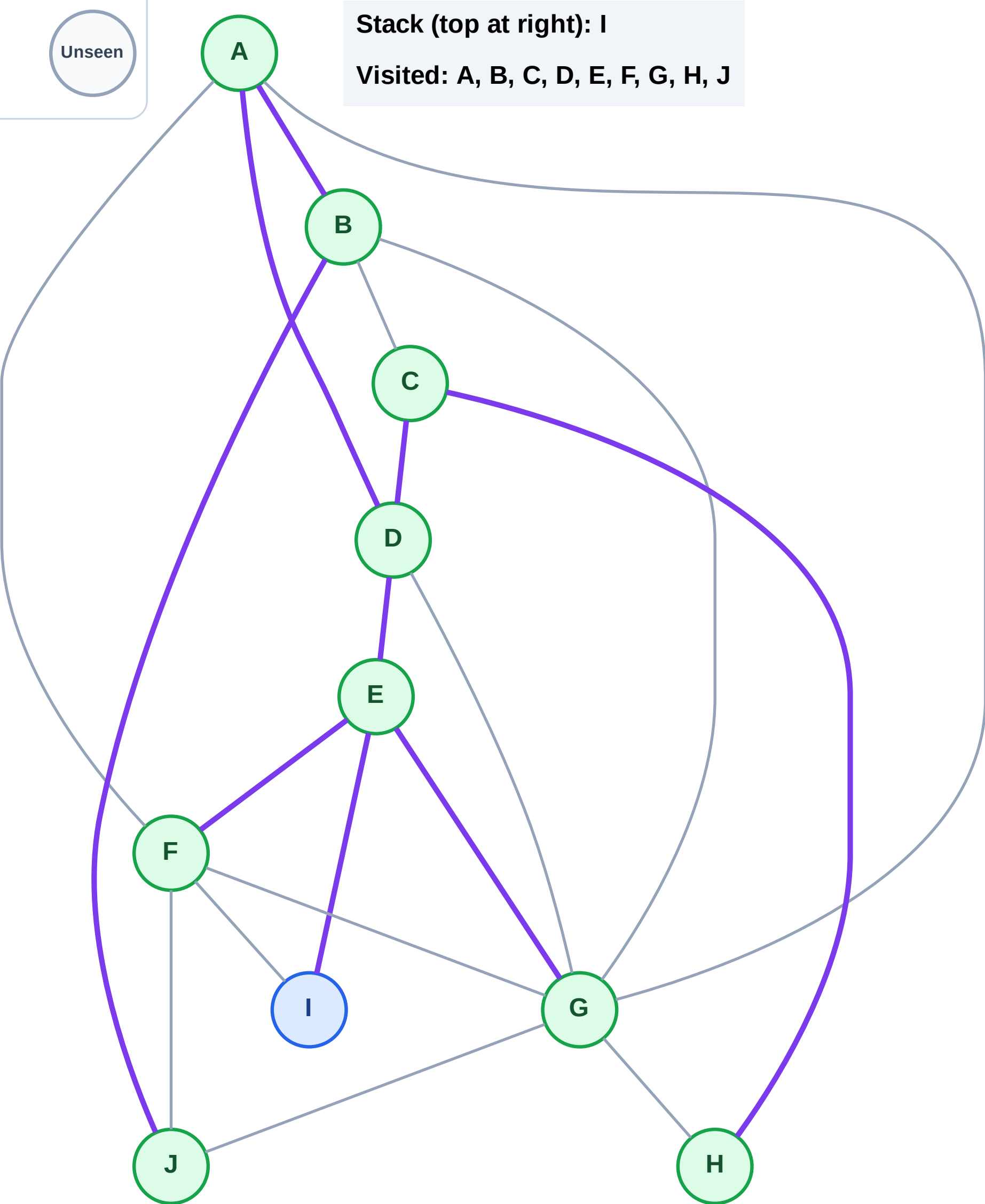
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

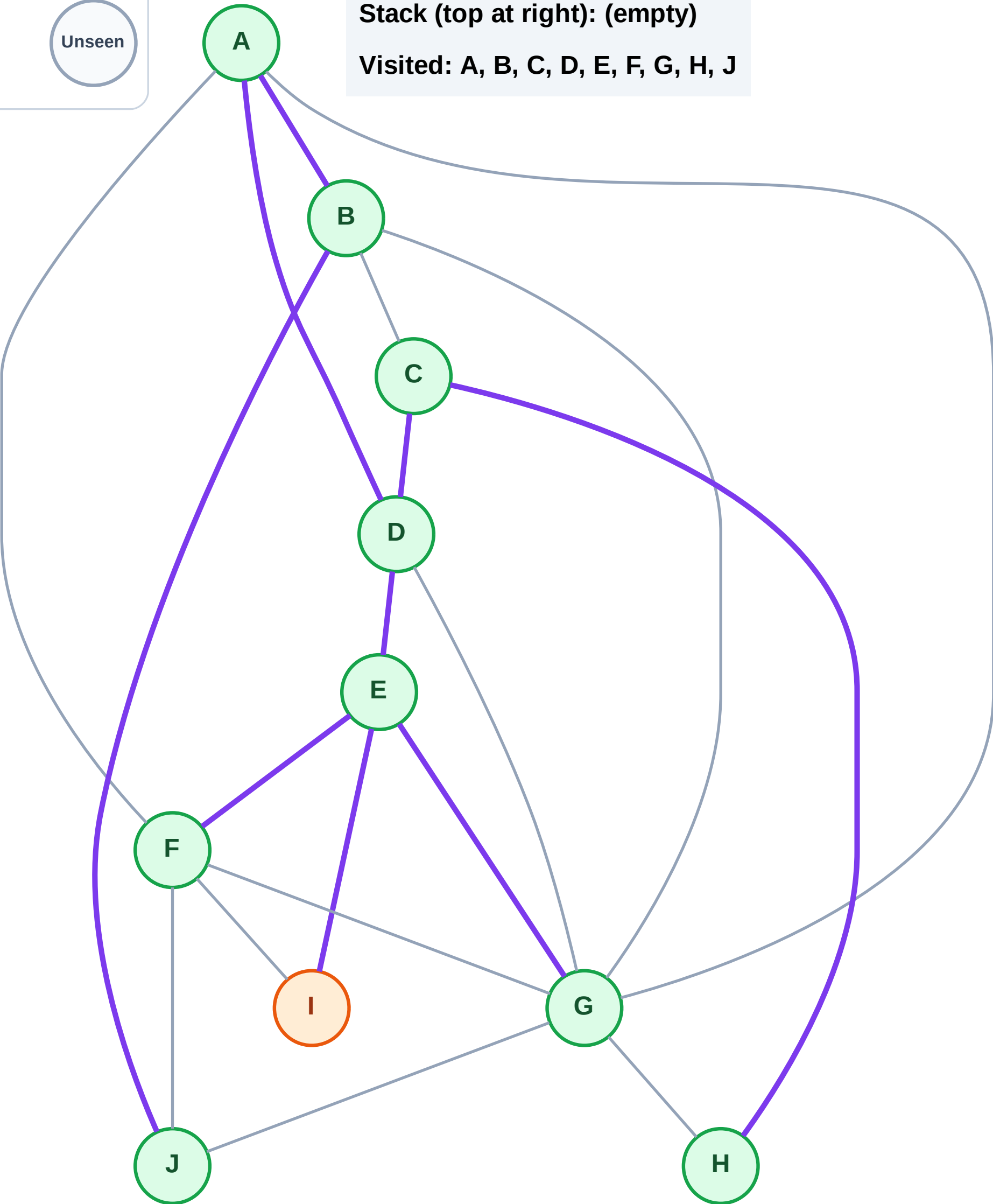
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

